Chapter 11
Response Surface Methods and Designs
Solutions

11.1S. A chemical plant produces oxygen by liquefying air and separating it into its component gases by fractional distillation. The purity of the oxygen is a function of the main condenser temperature and the pressure ratio between the upper and lower columns. Current operating conditions are temperature (ξ_1) = -220°C and pressure ratio (ξ_2) =1.2. Using the following data find the path of steepest ascent.

Temperature (ξ_1)	Pressure Ratio (ξ_2)	Purity
-225	1.1	82.8
-225	1.3	83.5
-215	1.1	84.7
-215	1.3	85.0
-220	1.2	84.1
-220	1.2	84.5
-220	1.2	83.9
-220	1.2	84.3

Design Expert Output

Response: Purity

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	3.14	2	1.57	26.17	0.0050	significant
A	*2.89*	*1*	*2.89*	*48.17*	*0.0023*	
B	*0.25*	*1*	*0.25*	*4.17*	*0.1108*	
Curvature	0.080	1	0.080	1.33	0.3125	not significant
Residual	0.24	4	0.060			
Lack of Fit	*0.040*	*1*	*0.040*	*0.60*	*0.4950*	*not significant*
Pure Error	*0.20*	*3*	*0.067*			
Cor Total	3.46	7				

The Model F-value of 26.17 implies the model is significant. There is only a 0.50% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.24	R-Squared	0.9290
Mean	84.10	Adj R-Squared	0.8935
C.V.	0.29	Pred R-Squared	0.7123
PRESS	1.00	Adeq Precision	12.702

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	84.00	1	0.12	83.66	84.34	
A-Temperature	0.85	1	0.12	0.51	1.19	1.00
B-Pressure Ratio	0.25	1	0.12	-0.090	0.59	1.00
Center Point	0.20	1	0.17	-0.28	0.68	1.00

Final Equation in Terms of Coded Factors:

Purity =
+84.00
+0.85 * A
+0.25 * B

Final Equation in Terms of Actual Factors:

Purity =
+118.40000
+0.17000 * Temperature

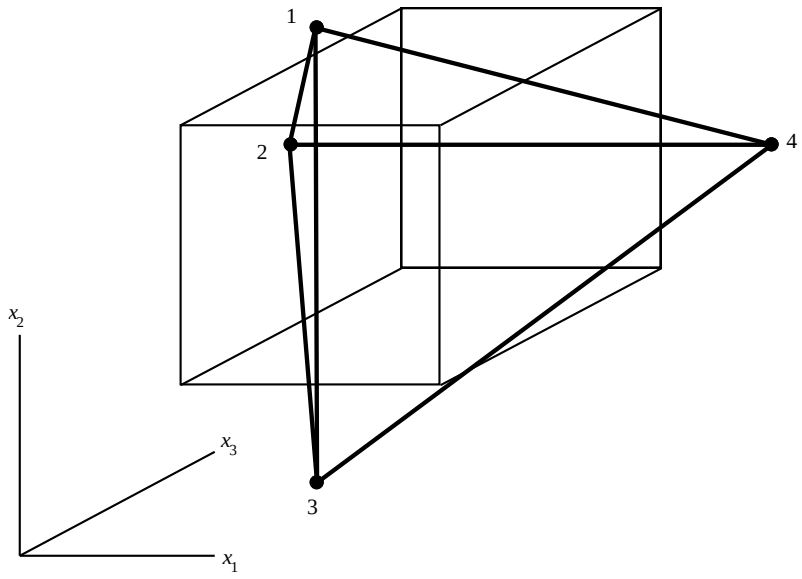
+2.50000 * Pressure Ratio

From the computer output use the model $\hat{y} = 84 + 0.85x_1 + 0.25x_2$as the equation for steepest ascent. Suppose we use a one degree change in temperature as the basic step size. Thus, the path of steepest ascent passes through the point (x_1=0, x_2=0) and has a slope 0.25/0.85. In the coded variables, one degree of temperature is equivalent to a step of $\Delta x_1 = 1/5 = 0.2$. Thus, $\Delta x_2 = (0.25/0.85)0.2 = 0.059$. The path of steepest ascent is:

	Coded Variables		Natural Variables	
	x_1	x_2	ξ_1	ξ_2
Origin	0	0	-220	1.2
Δ	0.2	0.059	1	0.0059
Origin +Δ	0.2	0.059	-219	1.2059
Origin +5Δ	1.0	0.295	-215	1.2295
Origin +7Δ	1.40	0.413	-213	1.2413

11.2. Verify that the following design is a simplex. Fit the first-order model and find the path of steepest ascent.

Position	x_1	x_2	x_3	y
1	0	$\sqrt{2}$	-1	18.5
2	$-\sqrt{2}$	0	1	19.8
3	0	$-\sqrt{2}$	-1	17.4
4	$\sqrt{2}$	0	1	22.5



The graphical representation of the design identifies a tetrahedron; therefore, the design is a simplex.

Design Expert Output

Response: y

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**
Model	14.49	3	4.83		
A	*3.64*	*1*	*3.64*		

B	*0.61*	*1*	*0.61*
C	*10.24*	*1*	*10.24*
Pure Error	0.000	0	
Cor Total	14.49	3	

Std. Dev.		R-Squared	1.0000
Mean	19.55	Adj R-Squared	
C.V.		Pred R-Squared	N/A
PRESS	N/A	Adeq Precision	0.000

Case(s) with leverage of 1.0000: Pred R-Squared and PRESS statistic not defined

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	19.55	1				
A-x1	1.35	1				1.00
B-x2	0.55	1				1.00
C-x3	1.60	1				1.00

Final Equation in Terms of Coded Factors:

y =
+19.55
+1.35 * A
+0.55 * B
+1.60 * C

Final Equation in Terms of Actual Factors:

y =
+19.55000
+0.95459 * x1
+0.38891 * x2
+1.60000 * x3

The first order model is $\hat{y} = 19.55 + 1.35x_1 + 0.55x_2 + 1.60x_3$.

To find the path of steepest ascent, let the basic step size be $\Delta x_3 = 1$. Then using the results obtained in the previous problem, we obtain

$$\Delta x_3 = \frac{\hat{\beta}_3}{2\lambda} \text{ or } 1.0 = \frac{1.60}{2\lambda}$$

which yields $2\lambda = 1.60$. Then the coordinates of points on the path of steepest ascent are defined by

$$\Delta x_1 = \frac{\hat{\beta}_1}{2\lambda} = 1.35/1.60 = 0.84$$
$$\Delta x_2 = \frac{\hat{\beta}_2}{2\lambda} = 0.55/1.60 = 0.34$$

Therefore, in the coded variables we have:

	Coded Variables		
	x_1	x_2	x_3
Origin	0	0	0
Δ	0.84	0.34	1.00
Origin +Δ	0.84	0.34	1.00
Origin +2	1.68	0.68	2.00

11.3. For the first-order model $\hat{y} = 60 + 1.5x_1 - 0.8x_2 + 2.0x_3$ find the path of steepest ascent. The variables are coded as $-1 \le x_i \le 1$.

Let the basic step size be $\Delta x_3 = 1$. $\Delta x_3 = \dfrac{\hat{\beta}_3}{2\lambda}$ or $1.0 = \dfrac{2.0}{2\lambda}$. Then $2\lambda = 2.0$

$$\Delta x_1 = \frac{\hat{\beta}_1}{2\lambda} = \frac{1.50}{2.0} = 0.75$$

$$\Delta x_2 = \frac{\hat{\beta}_2}{2\lambda} = \frac{-0.8}{2.0} = -0.40$$

Therefore, in the coded variables we have

	Coded Variables		
	x_1	x_2	x_3
Origin	0	0	0
Λ	0.75	-0.40	1.00
Origin +Δ	0.75	-0.40	1.00
Origin +2Δ	1.50	-0.80	2.00

11.4S. The region of experimentation for two factors are temperature $(100 \le T \le 300°\text{F})$ and catalyst feed rate $(10 \le C \le 30\text{lb/h})$. A first order model in the usual ± 1 coded variables has been fit to a molecular weight response, yielding the following model.

$$\hat{y} = 2000 + 125x_1 + 40x_2$$

(a) Find the path of steepest ascent.

$$x_1 = \frac{T - 200}{100} \qquad x_2 = \frac{C - 20}{10}$$

$$\Delta T = 100 \quad \Delta x_1 = \frac{100}{100} = 1$$

$$\Delta x_1 = \frac{\hat{\beta}_1}{2\lambda} \text{ or } 1 = \frac{125}{2\lambda} \text{ where } 2\lambda = 125$$

$$\Delta x_2 = \frac{\hat{\beta}_2}{2\lambda} = \frac{40}{125} = 0.32$$

	Coded Variables		Natural Variables	
	x_1	x_2	T	C
Origin	0	0	200	20
Λ	1	0.32	100	3.2
Origin +Δ	1	0.32	300	23.2
Origin +5Δ	5	1.60	700	36.0

(b) It is desired to move to a region where molecular weights are above 2500. Based on the information you have from the experimentation in this region, about how many steps along the path of steepest ascent might be required to move to the region of interest?

$$\Delta\hat{y} = \Delta x_1 \hat{\beta}_1 + \Delta x_2 \beta_2 = (1)(125) + (0.32)(40) = 137.8$$

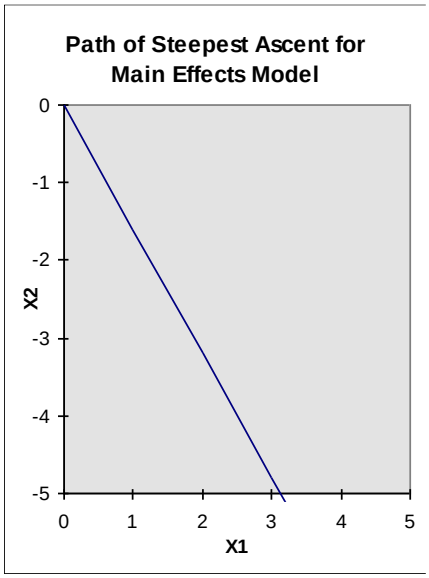
$$\#\,\text{Steps} = \frac{2500 - 2000}{137.8} = 3.63 \rightarrow 4$$

11.5. The path of steepest ascent is usually computed assuming that the model is truly first-order.; that is, there is no interaction. However, even if there is interaction, steepest ascent ignoring the interaction still usually produces good results. To illustrate, suppose that we have fit the model

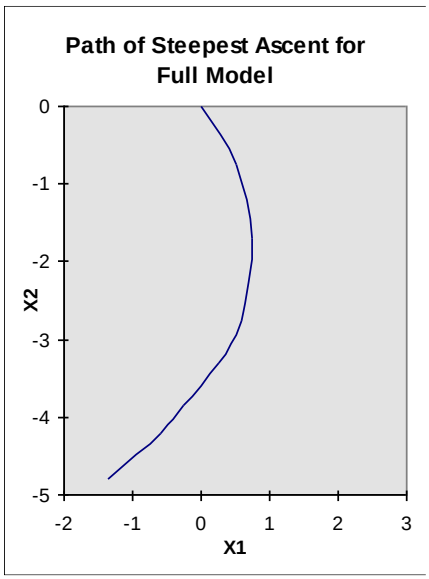
$$\hat{y} = 20 + 5x_1 - 8x_2 + 3x_1x_2$$

using coded variables ($-1 \le x_1 \le +1$)

(a) Draw the path of steepest ascent that you would obtain if the interaction were ignored.



(b) Draw the path of steepest ascent that you would obtain with the interaction included in the model. Compare this with the path found in part (a).



11.6. The data shown in Table P11.2 were collected in an experiment to optimize crystal growth as a function of three variables x_1, x_2, and x_3. Large values of y (yield in grams) are desirable. Fit a second order model and analyze the fitted surface. Under what set of conditions is maximum growth achieved?

Table P11.2

x_1	x_2	x_3	y
-1	-1	-1	66
-1	-1	1	70
-1	1	-1	78
-1	1	1	60
1	-1	-1	80
1	-1	1	70
1	1	-1	100
1	1	1	75
-1.682	0	0	100
1.682	0	0	80
0	-1.682	0	68
0	1.682	0	63
0	0	-1.682	65
0	0	1.682	82
0	0	0	113
0	0	0	100
0	0	0	118
0	0	0	88
0	0	0	100
0	0	0	85

Design Expert Output

Response: Yield

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	3662.00	9	406.89	2.19	0.1194	not significant
A	*22.08*	*1*	*22.08*	*0.12*	*0.7377*	
B	*25.31*	*1*	*25.31*	*0.14*	*0.7200*	
C	*30.50*	*1*	*30.50*	*0.16*	*0.6941*	
A^2	*204.55*	*1*	*204.55*	*1.10*	*0.3191*	
B^2	*2226.45*	*1*	*2226.45*	*11.96*	*0.0061*	
C^2	*1328.46*	*1*	*1328.46*	*7.14*	*0.0234*	
AB	*66.12*	*1*	*66.12*	*0.36*	*0.5644*	
AC	*55.13*	*1*	*55.13*	*0.30*	*0.5982*	
BC	*171.13*	*1*	*171.13*	*0.92*	*0.3602*	
Residual	1860.95	10	186.09			
Lack of Fit	*1001.61*	*5*	*200.32*	*1.17*	*0.4353*	*not significant*
Pure Error	*859.33*	*5*	*171.87*			
Cor Total	5522.95	19				

The "Model F-value" of 2.19 implies the model is not significant relative to the noise. There is a 11.94 % chance that a "Model F-value" this large could occur due to noise.

Std. Dev.	13.64	R-Squared	0.6631
Mean	83.05	Adj R-Squared	0.3598
C.V.	16.43	Pred R-Squared	-0.6034
PRESS	8855.23	Adeq Precision	3.882

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	100.67	1	5.56	88.27	113.06	
A-x1	1.27	1	3.69	-6.95	9.50	1.00
B-x2	1.36	1	3.69	-6.86	9.59	1.00
C-x3	-1.49	1	3.69	-9.72	6.73	1.00
A^2	-3.77	1	3.59	-11.77	4.24	1.02
B^2	-12.43	1	3.59	-20.44	-4.42	1.02
C^2	-9.60	1	3.59	-17.61	-1.59	1.02
AB	2.87	1	4.82	-7.87	13.62	1.00
AC	-2.63	1	4.82	-13.37	8.12	1.00
BC	-4.63	1	4.82	-15.37	6.12	1.00

Final Equation in Terms of Coded Factors:

$$
\begin{aligned}
\text{Yield} &= \\
&+100.67 \\
&+1.27 * A \\
&+1.36 * B \\
&-1.49 * C \\
&-3.77 * A^2 \\
&-12.43 * B^2 \\
&-9.60 * C^2 \\
&+2.87 * A * B \\
&-2.63 * A * C \\
&-4.63 * B * C
\end{aligned}
$$

Final Equation in Terms of Actual Factors:

$$
\begin{aligned}
\text{Yield} &= \\
&+100.66609 \\
&+1.27146 * x1 \\
&+1.36130 * x2 \\
&-1.49445 * x3 \\
&-3.76749 * x1^2 \\
&-12.42955 * x2^2 \\
&-9.60113 * x3^2
\end{aligned}
$$

+2.87500	* x1 * x2
-2.62500	* x1 * x3
-4.62500	* x2 * x3

There are so many non-significant terms in this model that we should consider eliminating some of them. A reasonable reduced model is shown below.

Design Expert Output

Response: Yield

ANOVA for Response Surface Reduced Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	3143.00	4	785.75	4.95	0.0095	significant
B	25.31	1	25.31	0.16	0.6952	
C	30.50	1	30.50	0.19	0.6673	
B2	2115.31	1	2115.31	13.33	0.0024	
C2	1239.17	1	1239.17	7.81	0.0136	
Residual	2379.95	15	158.66			
Lack of Fit	*1520.62*	*10*	*152.06*	*0.88*	*0.5953*	*not significant*
Pure Error	*859.33*	*5*	*171.87*			
Cor Total	5522.95	19				

The Model F-value of 4.95 implies the model is significant. There is only a 0.95% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	12.60	R-Squared	0.5691
Mean	83.05	Adj R-Squared	0.4542
C.V.	15.17	Pred R-Squared	0.1426
PRESS	4735.52	Adeq Precision	5.778

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	97.58	1	4.36	88.29	106.88	
B-x2	1.36	1	3.41	-5.90	8.63	1.00
C-x3	-1.49	1	3.41	-8.76	5.77	1.00
B2	-12.06	1	3.30	-19.09	-5.02	1.01
C2	-9.23	1	3.30	-16.26	-2.19	1.01

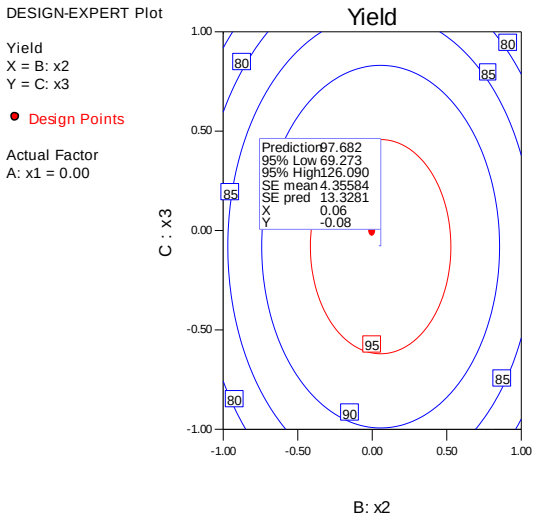
Final Equation in Terms of Coded Factors:

$$\begin{aligned} \text{Yield} = & \\ +97.58 & \\ +1.36 & * B \\ -1.49 & * C \\ -12.06 & * B^2 \\ -9.23 & * C^2 \end{aligned}$$

Final Equation in Terms of Actual Factors:

$$\begin{aligned} \text{Yield} = & \\ +97.58260 & \\ +1.36130 & * x2 \\ -1.49445 & * x3 \\ -12.05546 & * x2^2 \\ -9.22703 & * x3^2 \end{aligned}$$

The contour plot identifies a maximum near the center of the design space.



11.7. The data in Table P11.3 were collected by a chemical engineer. The response y is filtration time, x_1 is temperature, and x_2 is pressure. Fit a second-order model.

Table P11.3

x_1	x_2	y
-1	-1	54
-1	1	45
1	-1	32
1	1	47
-1.414	0	50
1.414	0	53
0	-1.414	47
0	1.414	51
0	0	41
0	0	39
0	0	44
0	0	42
0	0	40

Design Expert Output

Response: y

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	273.41	4	68.35	2.78	0.1018	not significant
A	*31.04*	*1*	*31.04*	*1.26*	*0.2937*	
B	*16.99*	*1*	*16.99*	*0.69*	*0.4299*	
A2	*81.39*	*1*	*81.39*	*3.31*	*0.1063*	
AB	*144.00*	*1*	*144.00*	*5.86*	*0.0418*	
Residual	196.59	8	24.57			
Lack of Fit	*181.79*	*4*	*45.45*	*12.28*	*0.0162*	*significant*
Pure Error	*14.80*	*4*	*3.70*			
Cor Total	470.00	12				

The "Model F-value" of 2.78 implies the model is not significant relative to the noise. There is a 10.18 % chance that a "Model F-value" this large could occur due to noise.

Std. Dev.	4.96	R-Squared	0.5817
Mean	45.00	Adj R-Squared	0.3726

C.V.	11.02	Pred R-Squared	-0.7171
PRESS	807.05	Adeq Precision	5.185

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	42.91	1	1.79	38.78	47.04	
A-Temperature	-2.79	1	2.48	-8.50	2.93	1.00
B-Pressure	2.06	1	2.48	-3.65	7.78	1.00
A2	6.78	1	3.73	-1.81	15.38	1.00
AB	12.00	1	4.96	0.57	23.43	1.00

Final Equation in Terms of Coded Factors:

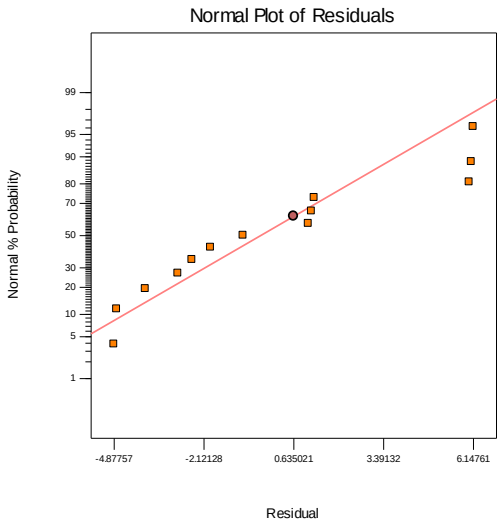
Time =
+42.91
+1.28 * A
-1.79 * B
+3.39 * A^2
+6.00 * A * B

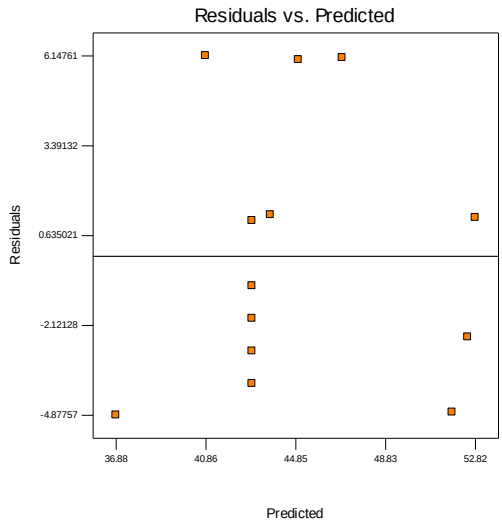
Time =
+42.91
-2.79 * A
+2.06 * B
+6.78 * A2
+12.00 * A * B

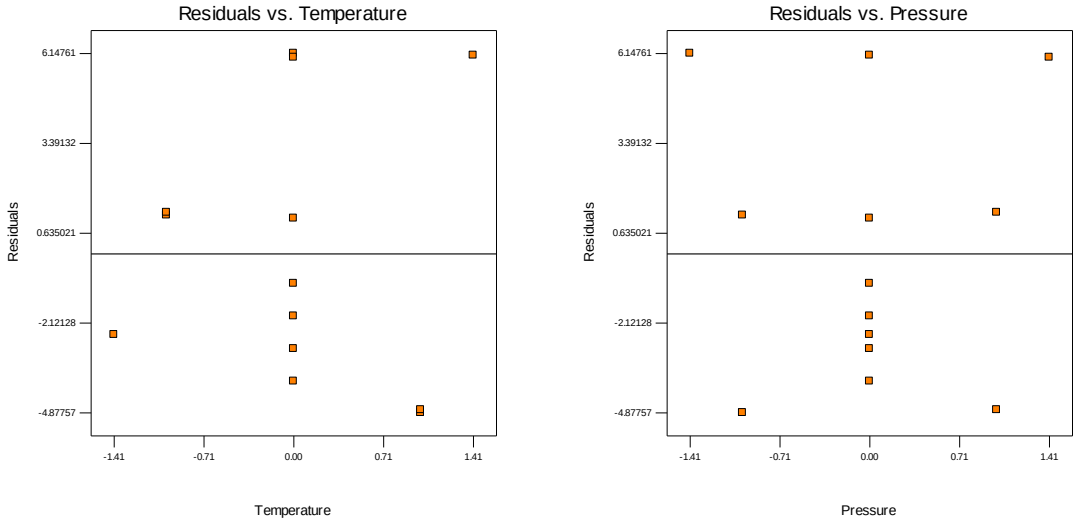
Final Equation in Terms of Actual Factors:

Time =
+42.91304
-1.96968 * Temperature
+1.45711 * Pressure
+3.39131 * Temperature2
+6.00000 * Temperature * Pressure

The lack of fit test in the above analysis is significant. The residual plots below are not unusual; however, the three largest residuals are axial points. Including cubic terms did not improve the lack of fit estimate and only made the PRESS larger.

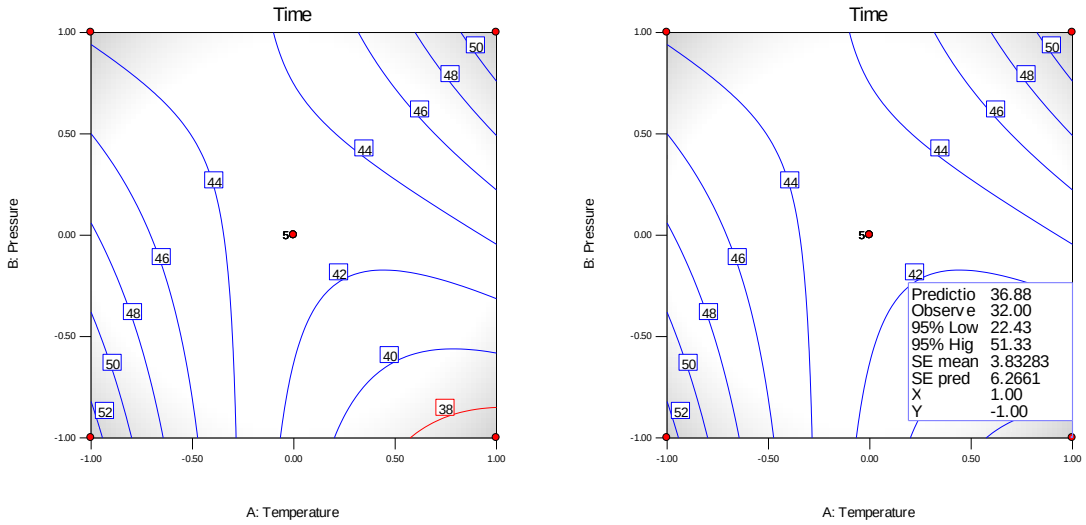






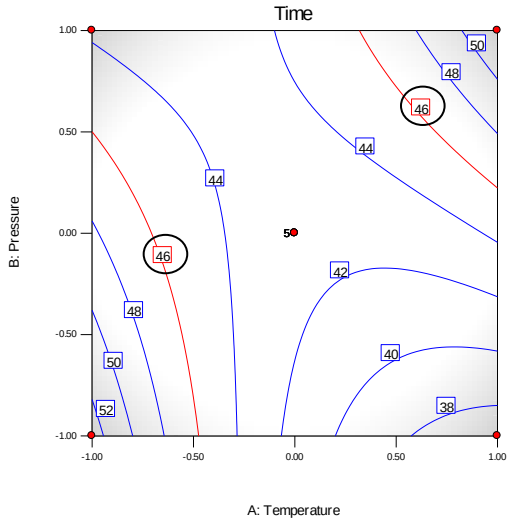
(a) What operating conditions would you recommend if the objective is to minimize the filtration time?

The contour plot below identifies the minimum filtration rate with the Temperature set at 1 and the Pressure at -1. By setting the flag at this location in *Design Expert*, the prediction of 36.88 filtration rate can be quickly estimated.



(b) What operating conditions would you recommend if the objective is to operate the process at a mean filtration time very close to 46?

The contour plot below identifies filtration time of 46 with the highlighted contours. There are two regions where a filtration rate of 46 is achievable. If higher temperature and pressure require higher operating costs, it would be preferential to choose the region with lower temperature and pressure.



11.8. An experimenter has run a Box-Behnken design and has obtained the results as shown in Table P11.5, where the response variable is the viscosity of a polymer.

Table P11.5

Level	Temp.	Agitation Rate	Pressure	x_1	x_2	x_3
High	200	10.0	25	+1	+1	+1
Middle	175	7.5	20	0	0	0
Low	150	5.0	15	-1	-1	-1

Run	x_1	x_2	x_3	y_1
1	-1	-1	0	535
2	1	-1	0	580
3	-1	1	0	596
4	1	1	0	563
5	-1	0	-1	645
6	1	0	-1	458
7	-1	0	1	350
8	1	0	1	600
9	0	-1	-1	595
10	0	1	-1	648
11	0	-1	1	532
12	0	1	1	656
13	0	0	0	653
14	0	0	0	599
15	0	0	0	620

(a) Fit the second-order model.

Design Expert Output

Response: Viscosity

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Model	89652.58	9	9961.40	9.54	0.0115	significant
A	*703.12*	*1*	*703.12*	*0.67*	*0.4491*	
B	*6105.12*	*1*	*6105.12*	*5.85*	*0.0602*	

C	*5408.00*	*1*	*5408.00*	*5.18*	*0.0719*	
A^2	*20769.23*	*1*	*20769.23*	*19.90*	*0.0066*	
B^2	*1404.00*	*1*	*1404.00*	*1.35*	*0.2985*	
C^2	*4719.00*	*1*	*4719.00*	*4.52*	*0.0868*	
AB	*1521.00*	*1*	*1521.00*	*1.46*	*0.2814*	
AC	*47742.25*	*1*	*47742.25*	*45.74*	*0.0011*	
BC	*1260.25*	*1*	*1260.25*	*1.21*	*0.3219*	
Residual	5218.75	5	1043.75			
Lack of Fit	*3736.75*	*3*	*1245.58*	*1.68*	*0.3941*	*not significant*
Pure Error	*1482.00*	*2*	*741.00*			
Cor Total	94871.33	14				

The Model F-value of 9.54 implies the model is significant. There is only a 1.15% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	32.31	R-Squared	0.9450
Mean	575.33	Adj R-Squared	0.8460
C.V.	5.62	Pred R-Squared	0.3347
PRESS	63122.50	Adeq Precision	10.425

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	624.00	1	18.65	576.05	671.95	
A-Temp	9.37	1	11.42	-19.99	38.74	1.00
B-Agitation Rate	27.62	1	11.42	-1.74	56.99	1.00
C-Pressure	-26.00	1	11.42	-55.36	3.36	1.00
A^2	-75.00	1	16.81	-118.22	-31.78	1.01
B^2	19.50	1	16.81	-23.72	62.72	1.01
C^2	-35.75	1	16.81	-78.97	7.47	1.01
AB	-19.50	1	16.15	-61.02	22.02	1.00
AC	109.25	1	16.15	67.73	150.77	1.00
BC	17.75	1	16.15	-23.77	59.27	1.00

Final Equation in Terms of Coded Factors:

Viscosity =
+624.00
+9.37 * A
+27.62 * B
-26.00 * C
-75.00 * A^2
+19.50 * B^2
-35.75 * C^2
-19.50 * A * B
+109.25 * A * C
+17.75 * B * C

Final Equation in Terms of Actual Factors:

Viscosity =
-629.50000
+27.23500 * Temp
-9.55000 * Agitation Rate
-111.60000 * Pressure
-0.12000 * Temp^2
+3.12000 * Agitation Rate^2
-1.43000 * Pressure^2
-0.31200 * Temp * Agitation Rate
+0.87400 * Temp * Pressure
+1.42000 * Agitation Rate * Pressure

(b) Perform the canonical analysis. What type of surface has been found?

Solution

Variable	Critical Value
*X*1	2.1849596
*X*2	-0.871371
*X*3	2.7586015
Predicted Value at Solution	586.34437

Eigenvalues and Eigenvectors

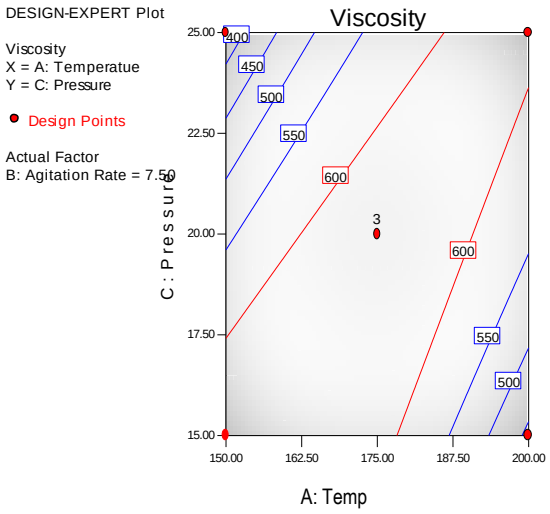
Variable	20.9229	2.5208	-114.694
*X*1	-0.02739	0.58118	0.81331
*X*2	0.99129	-0.08907	0.09703
*X*3	0.12883	0.80888	-0.57368

The system is a saddle point.

(c) What operating conditions on x_1, x_2, and x_3 lead to the stationary point?

The stationary point is (x_1, x_2, x_3) = (2.18496, -0.87167, 2.75860). This is outside the design region. It would be necessary to either examine contour plots or use numerical optimization methods to find desired operating conditions.

(d) What operating conditions would you recommend if it is important to obtain a viscosity that is as close to 600 as possible?



Any point on either of the contours showing a viscosity of 600 is satisfactory.

11.9S. Consider the three-variable central composite design shown in Table P11.6. Analyze the data and draw conclusions, assuming that we wish to maximize conversion (y_1) with activity (y_2) between 55 and 60.

Table P11.6

Run	Time (min)	Temperature (°C)	Catalyst (%)	Conversion (%) y_1	Activity y_2
1	-1.000	-1.000	-1.000	74.00	53.20
2	1.000	-1.000	-1.000	51.00	62.90
3	-1.000	1.000	-1.000	88.00	53.40
4	1.000	1.000	-1.000	70.00	62.60

5	-1.000	-1.000	1.000	71.00	57.30
6	1.000	-1.000	1.000	90.00	67.90
7	-1.000	1.000	1.000	66.00	59.80
8	1.000	1.000	1.000	97.00	67.80
9	0.000	0.000	0.000	81.00	59.20
10	0.000	0.000	0.000	75.00	60.40
11	0.000	0.000	0.000	76.00	59.10
12	0.000	0.000	0.000	83.00	60.60
13	-1.682	0.000	0.000	76.00	59.10
14	1.682	0.000	0.000	79.00	65.90
15	0.000	-1.682	0.000	85.00	60.00
16	0.000	1.682	0.000	97.00	60.70
17	0.000	0.000	-1.682	55.00	57.40
18	0.000	0.000	1.682	81.00	63.20
19	0.000	0.000	0.000	80.00	60.80
20	0.000	0.000	0.000	91.00	58.90

Quadratic models are developed for the Conversion and Activity response variables as follows:

Design Expert Output

Response: Conversion

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	2555.73	9	283.97	12.76	0.0002	significant
A	*14.44*	*1*	*14.44*	*0.65*	*0.4391*	
B	*222.96*	*1*	*222.96*	*10.02*	*0.0101*	
C	*525.64*	*1*	*525.64*	*23.63*	*0.0007*	
A^2	*48.47*	*1*	*48.47*	*2.18*	*0.1707*	
B^2	*124.48*	*1*	*124.48*	*5.60*	*0.0396*	
C^2	*388.59*	*1*	*388.59*	*17.47*	*0.0019*	
AB	*36.13*	*1*	*36.13*	*1.62*	*0.2314*	
AC	*1035.13*	*1*	*1035.13*	*46.53*	*< 0.0001*	
BC	*120.12*	*1*	*120.12*	*5.40*	*0.0425*	
Residual	222.47	10	22.25			
Lack of Fit	*56.47*	*5*	*11.29*	*0.34*	*0.8692*	*not significant*
Pure Error	*166.00*	*5*	*33.20*			
Cor Total	287.28	19				

The Model F-value of 12.76 implies the model is significant. There is only a 0.02% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	4.72	R-Squared	0.9199
Mean	78.30	Adj R-Squared	0.8479
C.V.	6.02	Pred R-Squared	0.7566
PRESS	676.22	Adeq Precision	14.239

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	81.09	1	1.92	76.81	85.38	
A-Time	1.03	1	1.28	-1.82	3.87	1.00
B-Temperature	4.04	1	1.28	1.20	6.88	1.00
C-Catalyst	6.20	1	1.28	3.36	9.05	1.00
A2	-1.83	1	1.24	-4.60	0.93	1.02
B2	2.94	1	1.24	0.17	5.71	1.02
C2	-5.19	1	1.24	-7.96	-2.42	1.02
AB	2.13	1	1.67	-1.59	5.84	1.00
AC	11.38	1	1.67	7.66	15.09	1.00
BC	-3.87	1	1.67	-7.59	-0.16	1.00

Final Equation in Terms of Coded Factors:

Conversion =
+81.09
+1.03 * A

+4.04 * B
+6.20 * C
-1.83 * A2
+2.94 * B2
-5.19 * C2
+2.13 * A * B
+11.38 * A * C
-3.87 * B * C

Final Equation in Terms of Actual Factors:

Conversion =
+81.09128
+1.02845 * Time
+4.04057 * Temperature
+6.20396 * Catalyst
-1.83398 * Time2
+2.93899 * Temperature2
-5.19274 * Catalyst2
+2.12500 * Time * Temperature
+11.37500 * Time * Catalyst
-3.87500 * Temperature * Catalyst

Design Expert Output

Response: Activity

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	256.20	9	28.47	9.16	0.0009	significant
A	*175.35*	*1*	*175.35*	*56.42*	*< 0.0001*	
B	*0.89*	*1*	*0.89*	*0.28*	*0.6052*	
C	*67.91*	*1*	*67.91*	*21.85*	*0.0009*	
A^2	*10.05*	*1*	*10.05*	*3.23*	*0.1024*	
B^2	*0.081*	*1*	*0.081*	*0.026*	*0.8753*	
C^2	*0.047*	*1*	*0.047*	*0.015*	*0.9046*	
AB	*1.20*	*1*	*1.20*	*0.39*	*0.5480*	
AC	*0.011*	*1*	*0.011*	*3.620E-003*	*0.9532*	
BC	*0.78*	*1*	*0.78*	*0.25*	*0.6270*	
Residual	31.08	10	3.11			
Lack of Fit	*27.43*	*5*	*5.49*	*7.51*	*0.0226*	*significant*
Pure Error	*3.65*	*5*	*0.73*			
Cor Total	287.28	19				

The Model F-value of 9.16 implies the model is significant. There is only a 0.09% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	1.76	R-Squared	0.8918
Mean	60.51	Adj R-Squared	0.7945
C.V.	2.91	Pred R-Squared	0.2536
PRESS	214.43	Adeq Precision	10.911

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	59.85	1	0.72	58.25	61.45	
A-Time	3.58	1	0.48	2.52	4.65	1.00
B-Temperature	0.25	1	0.48	-0.81	1.32	1.00
C-Catalyst	2.23	1	0.48	1.17	3.29	1.00
A^2	0.83	1	0.46	-0.20	1.87	1.02
B^2	0.075	1	0.46	-0.96	1.11	1.02
C^2	0.057	1	0.46	-0.98	1.09	1.02
AB	-0.39	1	0.62	-1.78	1.00	1.00
AC	-0.038	1	0.62	-1.43	1.35	1.00
BC	0.31	1	0.62	-1.08	1.70	1.00

Final Equation in Terms of Coded Factors:

Conversion =
+59.85

+3.58 * A
+0.25 * B
+2.23 * C
+0.83 * A^2
+0.075 * B^2
+0.057 * C^2
-0.39 * A * B
-0.038 * A * C
+0.31 * B * C

Final Equation in Terms of Actual Factors:

Conversion =
+59.84984
+3.58327 * Time
+0.25462 * Temperature
+2.22997 * Catalyst
+0.83491 * $Time^2$
+0.074772 * $Temperature^2$
+0.057094 * $Catalyst^2$
-0.38750 * Time * Temperature
-0.037500 * Time * Catalyst
+0.31250 * Temperature * Catalyst

Because many of the terms are insignificant, the reduced quadratic model is fit as follows:

Design Expert Output

Response: Activity

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	253.20	3	84.40	39.63	< 0.0001	significant
A	*175.35*	*1*	*175.35*	*82.34*	*< 0.0001*	
C	*67.91*	*1*	*67.91*	*31.89*	*< 0.0001*	
A^2	*9.94*	*1*	*9.94*	*4.67*	*0.0463*	
Residual	34.07	16	2.13			
Lack of Fit	*30.42*	*11*	*2.77*	*3.78*	*0.0766*	*not significant*
Pure Error	*3.65*	*5*	*0.73*			
Cor Total	287.28	19				

The Model F-value of 39.63 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	1.46	R-Squared	0.8814
Mean	60.51	Adj R-Squared	0.8591
C.V.	2.41	Pred R-Squared	0.6302
PRESS	106.24	Adeq Precision	20.447

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	59.95	1	0.42	59.06	60.83	
A-Time	3.58	1	0.39	2.75	4.42	1.00
C-Catalyst	2.23	1	0.39	1.39	3.07	1.00
A^2	0.82	1	0.38	0.015	1.63	1.00

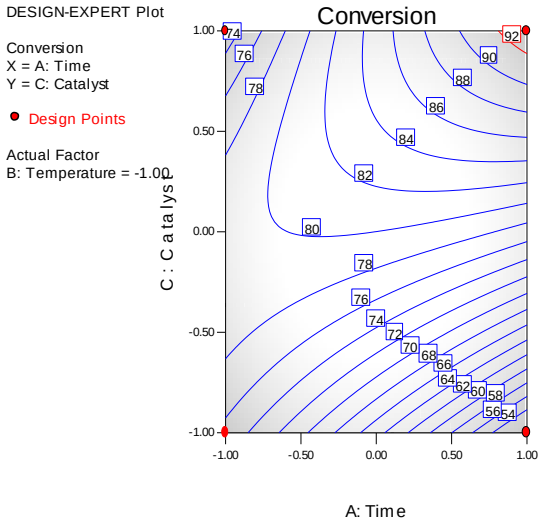
Final Equation in Terms of Coded Factors:

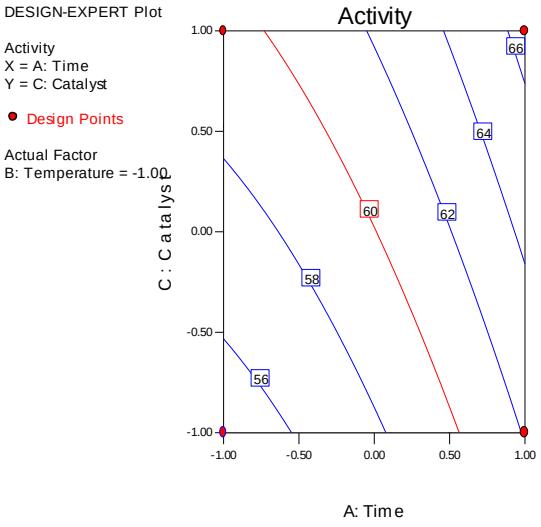
Activity =
+59.95
+3.58 * A
+2.23 * C
+0.82 * A2

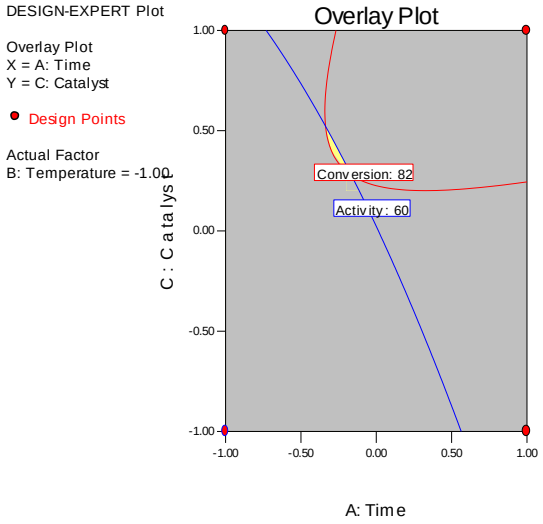
Final Equation in Terms of Actual Factors:

Activity =

+59.94802
+3.58327 * Time
+2.22997 * Catalyst
+0.82300 * Time2







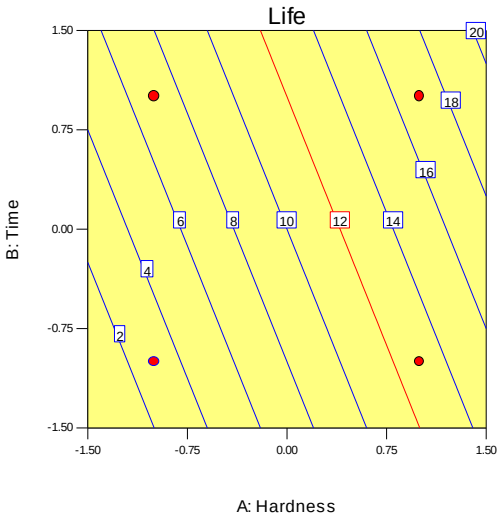
The contour plots visually describe the models while the overlay plots identifies the acceptable region for the process.

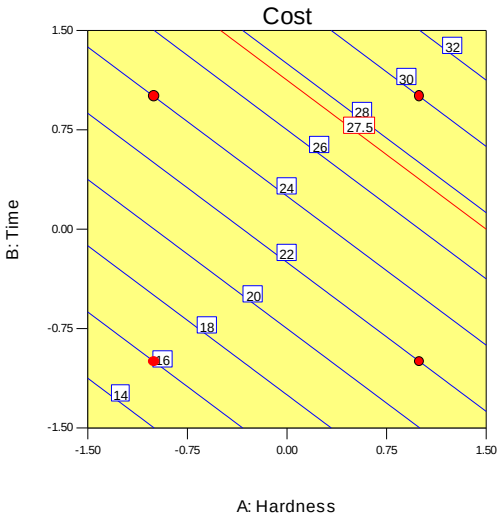
11.10. A manufacturer of cutting tools has developed two empirical equations for tool life in hours (y_1) and for tool cost in dollars (y_2). Both models are linear functions of steel hardness (x_1) and manufacturing time (x_2). The two equations are

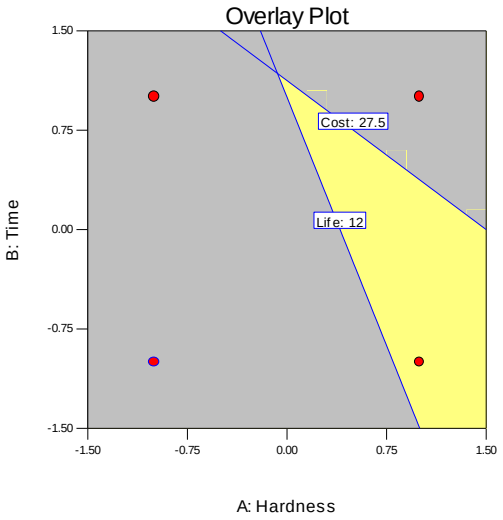
$$\hat{y}_1 = 10 + 5x_1 + 2x_2$$
$$\hat{y}_2 = 23 + 3x_1 + 4x_2$$

and both equations are valid over the range $-1.5 \le x_i \le 1.5$. Unit tool cost must be below \$27.50 and life must exceed 12 hours for the product to be competitive. Is there a feasible set of operating conditions for this process? Where would you recommend that the process be run?

The contour plots below graphically describe the two models. The overlay plot identifies the feasible operating region for the process.







$$10 + 5x_1 + 2x_2 \ge 12$$
$$23 + 3x_1 + 4x_2 \le 27.50$$

11.11. A central composite design is run in a chemical vapor deposition process, resulting in the experimental data shown in Table P11.7. Four experimental units were processed simultaneously on each run of the design, and the responses are the mean and variance of thickness, computed across the four units.

Table P11.7

x_1	x_2	$\bar{y}$	s^2
-1	-1	360.6	6.689
1	-1	445.2	14.230
-1	1	412.1	7.088
1	1	601.7	8.586
1.414	0	518.0	13.130
-1.414	0	411.4	6.644
0	1.414	497.6	7.649
0	-1.414	397.6	11.740
0	0	530.6	7.836
0	0	495.4	9.306
0	0	510.2	7.956
0	0	487.3	9.127

(a) Fit a model to the mean response. Analyze the residuals.

Design Expert Output

Response: Mean Thick

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	47644.26	5	9528.85	16.12	0.0020	significant
A	*22573.36*	*1*	*22573.36*	*38.19*	*0.0008*	
B	*15261.91*	*1*	*15261.91*	*25.82*	*0.0023*	
A^2	*2795.58*	*1*	*2795.58*	*4.73*	*0.0726*	
B^2	*5550.74*	*1*	*5550.74*	*9.39*	*0.0221*	
AB	*2756.25*	*1*	*2756.25*	*4.66*	*0.0741*	
Residual	3546.83	6	591.14			
Lack of Fit	*2462.04*	*3*	*820.68*	*2.27*	*0.2592*	*not significant*
Pure Error	*1084.79*	*3*	*361.60*			
Cor Total	51191.09	11				

The Model F-value of 16.12 implies the model is significant. There is only a 0.20% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	24.31	R-Squared	0.9307
Mean	472.31	Adj R-Squared	0.8730
C.V.	5.15	Pred R-Squared	0.6203
PRESS	19436.37	Adeq Precision	11.261

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	505.88	1	12.16	476.13	535.62	
A-x1	53.12	1	8.60	32.09	74.15	1.00
B-x2	43.68	1	8.60	22.64	64.71	1.00
A^2	-20.90	1	9.61	-44.42	2.62	1.04
B^2	-29.45	1	9.61	-52.97	-5.93	1.04
AB	26.25	1	12.16	-3.50	56.00	1.00

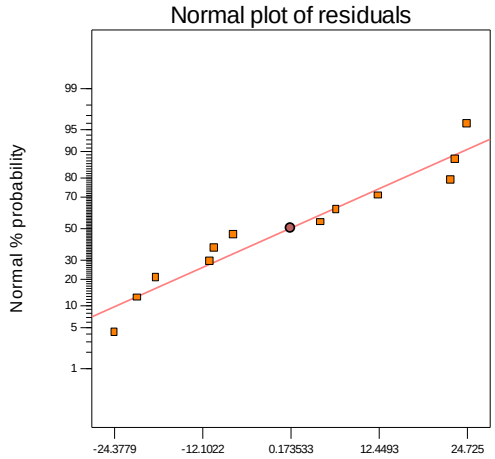
Final Equation in Terms of Coded Factors:

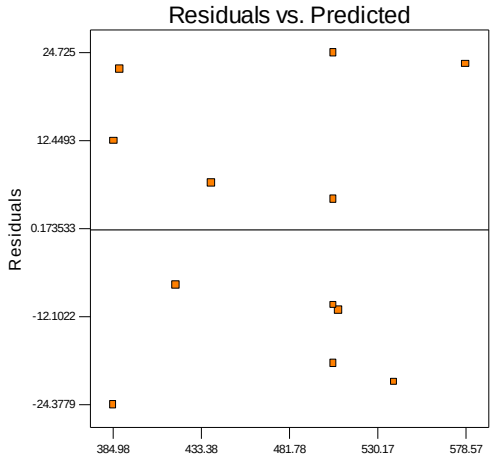
$$\text{Mean Thick} = +505.88 + 53.12 * A + 43.68 * B - 20.90 * A^2 - 29.45 * B^2$$

+26.25 * A * B

Final Equation in Terms of Actual Factors:

Mean Thick =
+505.87500
+53.11940 * x1
+43.67767 * x2
-20.90000 * $x1^2$
-29.45000 * $x2^2$
+26.25000 * x1 * x2





A modest deviation from normality can be observed in the Normal Plot of Residuals; however, not enough to be concerned.

(b) Fit a model to the variance response. Analyze the residuals.

Design Expert Output

Response: Var Thick

ANOVA for Response Surface 2FI Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	65.80	3	21.93	35.86	< 0.0001	significant
A	*41.46*	*1*	*41.46*	*67.79*	*< 0.0001*	
B	*15.21*	*1*	*15.21*	*24.87*	*0.0011*	
AB	*9.13*	*1*	*9.13*	*14.93*	*0.0048*	
Residual	4.89	8	0.61			
Lack of Fit	*3.13*	*5*	*0.63*	*1.06*	*0.5137*	*not significant*
Pure Error	*1.77*	*3*	*0.59*			
Cor Total	70.69	11				

The Model F-value of 35.86 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.78	R-Squared	0.9308
Mean	9.17	Adj R-Squared	0.9048
C.V.	8.53	Pred R-Squared	0.8920
PRESS	7.64	Adeq Precision	18.572

Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	9.17	1	0.23	8.64	9.69	

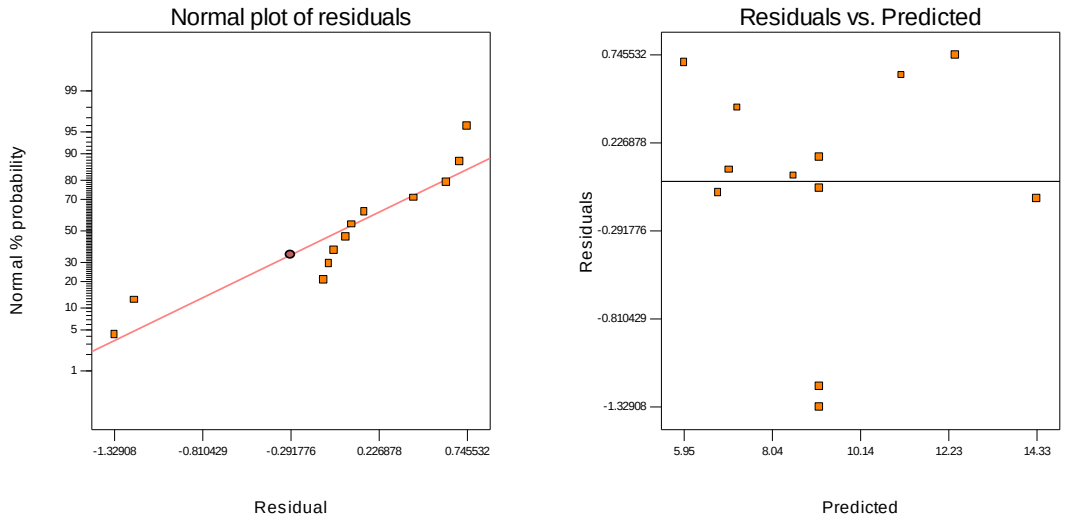
A-x1	2.28	1	0.28	1.64	2.91	1.00
B-x2	-1.38	1	0.28	-2.02	-0.74	1.00
AB	-1.51	1	0.39	-2.41	-0.61	1.00

Final Equation in Terms of Coded Factors:

Var Thick =
+9.17
+2.28 * A
-1.38 * B
-1.51 * A * B

Final Equation in Terms of Actual Factors:

Var Thick =
+9.16508
+2.27645 * x1
-1.37882 * x2
-1.51075 * x1 * x2



The residual plots are not acceptable. A transformation should be considered. If not successful at correcting the residual plots, further investigation into the two apparently unusual points should be made.

(c) Fit a model to the $\ln(s^2)$. Is this model superior to the one you found in part (b)?

Design Expert Output

Response: Var Thick Transform: Natural log Constant: 0

ANOVA for Response Surface 2FI Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	0.67	3	0.22	36.94	< 0.0001	significant
A	*0.46*	*1*	*0.46*	*74.99*	*< 0.0001*	
B	*0.14*	*1*	*0.14*	*22.80*	*0.0014*	
AB	*0.079*	*1*	*0.079*	*13.04*	*0.0069*	
Residual	0.049	8	6.081E-003			
Lack of Fit	*0.024*	*5*	*4.887E-003*	*0.61*	*0.7093*	*not significant*
Pure Error	*0.024*	*3*	*8.071E-003*			
Cor Total	0.72	11				

The Model F-value of 36.94 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev. 0.078 R-Squared 0.9327

Mean	2.18	Adj R-Squared	0.9074
C.V.	3.57	Pred R-Squared	0.8797
PRESS	0.087	Adeq Precision	18.854

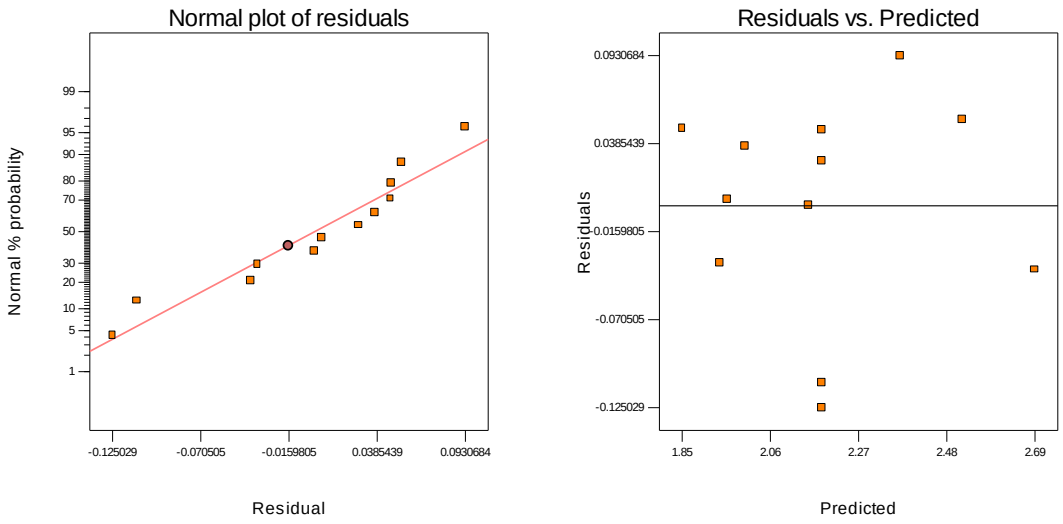
Factor	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High	VIF
Intercept	2.18	1	0.023	2.13	2.24	
A-x1	0.24	1	0.028	0.18	0.30	1.00
B-x2	-0.13	1	0.028	-0.20	-0.068	1.00
AB	-0.14	1	0.039	-0.23	-0.051	1.00

Final Equation in Terms of Coded Factors:

Ln(Var Thick) =
+2.18
+0.24 * A
-0.13 * B
-0.14 * A * B

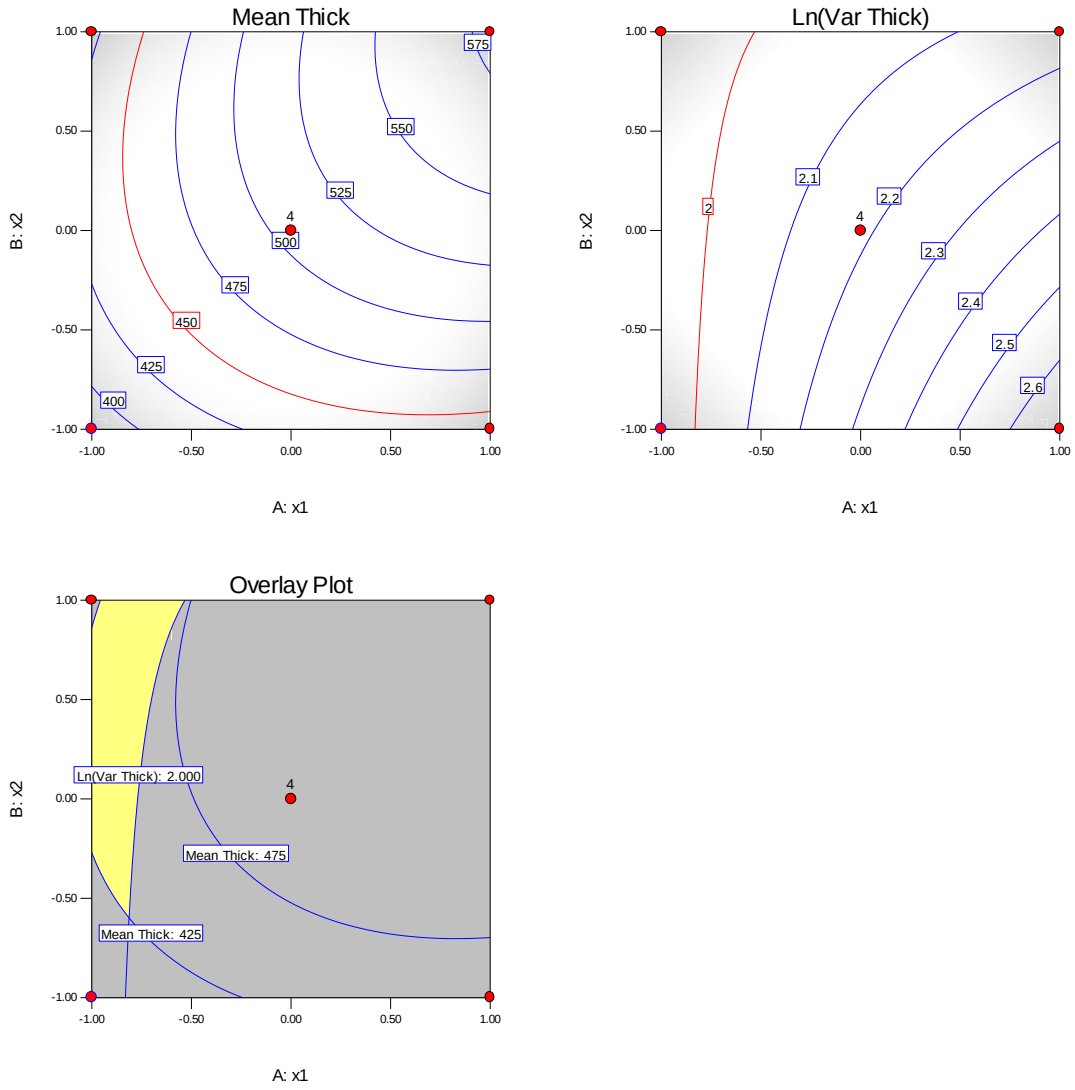
Final Equation in Terms of Actual Factors:

Ln(Var Thick) =
+2.18376
+0.23874 * x1
-0.13165 * x2
-0.14079 * x1 * x2



The residual plots are much improved following the natural log transformation; however, the two runs still appear to be somewhat unusual and should be investigated further. They will be retained in the analysis.

(d) Suppose you want the mean thickness to be in the interval 450 ± 25. Find a set of operating conditions that achieve the objective and simultaneously minimize the variance.



The contour plots describe the two models while the overlay plot identifies the acceptable region for the process.

(e) Discuss the variance minimization aspects of part (d). Have you minimized total process variance?

The within run variance has been minimized; however, the run-to-run variation has not been minimized in the analysis. This may not be the most robust operating conditions for the process.

11.12S. Verify that an orthogonal first-order design is also first-order rotatable.

To show that a first order orthogonal design is also first order rotatable, consider

$$V(\hat{y}) = V(\hat{\beta}_0 + \sum_{i=1}^{k} \beta_i x_i) = V(\hat{\beta}_0) + \sum_{i=1}^{k} x_i^2 V(\beta_i)$$

since all covariances between $\hat{\beta}_i$ and $\hat{\beta}_j$ are zero, due to design orthogonality. Furthermore, we have:

$$V\left(\hat{\beta}_0\right) = V(\beta_1) = V(\hat{\beta}_2) = \ldots = V(\beta_k) = \frac{\sigma^2}{n}, \text{ so}$$

$$V(\hat{y}) = \frac{\sigma^2}{n} + \frac{\sigma^2}{n}\sum_{i=1}^{k} x_i^2$$

$$V(\hat{y}) = \frac{\sigma^2}{n}\left(1 + \sum_{i=1}^{k} x_i^2\right)$$

which is a function of distance from the design center (i.e. $\mathbf{x}=\mathbf{0}$), and not direction. Thus the design must be rotatable. Note that n is, in general, the number of points in the exterior portion of the design. If there are n_c center points, then $V(\hat{\beta}_0) = \frac{\sigma^2}{(n+n_c)}$.

11.13. Show that augmenting a 2^k design with n_c center points does not affect the estimates of the β_i (i=1, 2, . . . , k), but that the estimate of the intercept β_0 is the average of all $2^k + n_c$ observations.

In general, the **X** matrix for the 2^k design with n_c center points and the **y** vector would be:

$$\mathbf{X} = \begin{array}{c} \begin{matrix} \beta_0 & \beta_0 & \beta_0 & \cdots & \beta_0 \end{matrix} \\ \left[\begin{array}{ccccc} 1 & -1 & -1 & \cdots & -1 \\ 1 & 1 & -1 & \cdots & -1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \\ \hline 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{array}\right] \end{array}$$

← The upper half of the matrix is the usual ± 1 notation of the 2^k design

← The lower half of the matrix represents the center points (n_c rows)

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{2^k} \\ \hline n_{0_1} \\ n_{0_2} \\ \vdots \\ n_{0_c} \end{bmatrix}$$ ← 2^k+n_c observations.

$$\mathbf{X'X} = \begin{bmatrix} 2^k + n_c & 0 & \cdots & 0 \\ & 2^k & \cdots & 0 \\ & & \ddots & \vdots \\ & & & 2^k \end{bmatrix} \qquad \mathbf{X'y} = \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ \vdots \\ g_k \end{bmatrix}$$

← Grand total of all 2^k+n_c observations

← usual contrasts from 2^k

Therefore, $\hat{\beta}_0 = \frac{g_0}{2^k + n_c}$, which is the average of all $\left(2^k + n_c\right)$observations, while $\hat{\beta}_i = \frac{g_i}{2^k}$, which does not depend on the number of center points, since in computing the contrasts g_i, all observations at the center are multiplied by zero.

11.14S. Verify that the central composite design shown in Table P11.8 blocks orthogonally.

Table P11.8

Block 1			Block 2			Block 3		
x_1	x_2	x_3	x_1	x_2	x_3	x_1	x_2	x_3
0	0	0	0	0	0	-1.633	0	0
0	0	0	0	0	0	1.633	0	0
1	1	1	1	1	-1	0	-1.633	0
1	-1	-1	1	-1	1	0	1.633	0
-1	-1	1	-1	1	1	0	0	-1.633
-1	1	-1	-1	-1	-1	0	0	1.633
						0	0	0
						0	0	0

Note that each block is an orthogonal first order design, since the cross products of elements in different columns add to zero for each block. To verify the second condition, choose a column, say column x_2. Now

$$\sum_{u=1}^{k} x_{2u}^2 = 13.334, \text{ and } n = 20$$

For blocks 1 and 2,

$$\sum_m x_{2m}^2 = 4,\ n_m = 6,\ \text{so}$$

$$\frac{\sum_m x_{2m}^2}{\sum_{u=1}^n x_{2u}^2} = \frac{n_m}{n}$$

$$\frac{4}{13.334} = \frac{6}{20}$$

$$0.3 = 0.3$$

and condition 2 is satisfied by blocks 1 and 2. For block 3, we have

$$\sum_m x_{2m}^2 = 5.334,\ n_m = 8,\ \text{so}$$

$$\frac{\sum_m x_{2m}^2}{\sum_{u=1}^n x_{2u}^2} = \frac{n_m}{n}$$

$$\frac{5.334}{13.334} = \frac{8}{20}$$

$$0.4 = 0.4$$

And condition 2 is satisfied by block 3. Similar results hold for the other columns.

11.15. ***Blocking in the central composite design***. Consider a central composite design for $k = 4$ variables in two blocks. Can a rotatable design always be found that blocks orthogonally?

To run a central composite design in two blocks, assign the n_f factorial points and the n_{c1} center points to block 1 and the $2k$ axial points plus n_{c2} center points to block 2. Both blocks will be orthogonal first order designs, so the first condition for orthogonal blocking is satisfied.

The second condition implies that

$$\frac{\sum_m x_{im}^2\,(block1)}{\sum_m x_{im}^2\,(block2)} = \frac{n_f + n_{c1}}{2k + n_{c2}}$$

However, $\sum_m x_{im}^2 = n_f$ in block 1 and $\sum_m x_{im}^2 = 2\alpha^2$ in block 2, so

$$\frac{n_f}{2\alpha^2} = \frac{n_f + n_{c1}}{2k + n_{c2}}$$

Which gives:

$$\alpha = \left[\frac{n_f \left(2k + n_{c2}\right)}{2\left(n_f + n_{c1}\right)} \right]^{\frac{1}{2}}$$

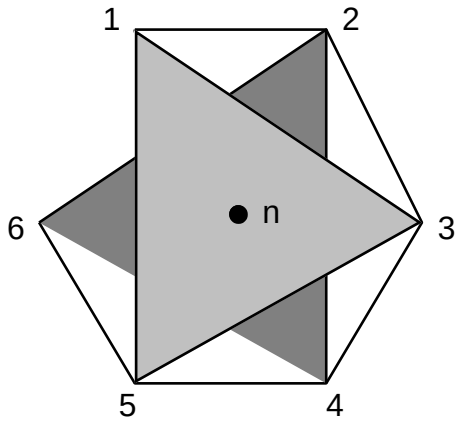
Since $\alpha = \sqrt[4]{n_f}$ if the design is to be rotatable, then the design must satisfy

$$n_f = \left[\frac{n_f \left(2k + n_{c2}\right)}{2\left(n_f + n_{c1}\right)} \right]^2$$

It is not possible to find rotatable central composite designs which block orthogonally for all *k*. For example, if *k*=3, the above condition cannot be satisfied. For *k*=2, there must be an equal number of center points in each block, i.e. $n_{c1} = n_{c2}$. For *k*=4, we must have $n_{c1} = 4$ and $n_{c2} = 2$.

11.16. How could a hexagon design be run in two orthogonal blocks?

The hexagonal design can be blocked as shown below. There are $n_{c1} = n_{c2} = n_c$ center points with n_c even.



Put the points 1,3,and 5 in block 1 and 2,4,and 6 in block 2. Note that each block is a simplex.

11.17. Yield during the first four cycles of a chemical process is shown in the following table. The variables are percent concentration (x_1) at levels 30, 31, and 32 and temperature (x_2) at 140, 142, and 144°F. Analyze by EVOP methods.

	Conditions				
Cycle	(1)	(2)	(3)	(4)	(5)
1	60.7	59.8	60.2	64.2	57.5
2	59.1	62.8	62.5	64.6	58.3
3	56.6	59.1	59.0	62.3	61.1
4	60.5	59.8	64.5	61.0	60.1

Cycle: n=1 Phase 1

Calculation of Averages							Calculation of Standard Deviation
Operating Conditions		(1)	(2)	(3)	(4)	(5)	
(i)	Previous Cycle Sum						Previous Sum S=
(ii)	Previous Cycle Average						Previous Average =
(iii)	New Observation	60.7	59.8	60.2	64.2	57.5	New S=Range x $f_{k,n}$
(iv)	Differences						Range=
(v)	New Sums	60.7	59.8	60.2	64.2	57.5	New Sum S=
(vi)	New Averages	60.7	59.8	60.2	64.2	57.5	New average S = New Sum S/(n-1)=

Calculation of Effects	Calculation of Error Limits
$A = \frac{1}{2}(\bar{y}_3 + \bar{y}_4 - \bar{y}_2 - \bar{y}_5) = 3.55$	For New Average: $\left(\frac{2}{\sqrt{n}}\right)S =$
$B = \frac{1}{2}(\bar{y}_3 - \bar{y}_4 - \bar{y}_2 + \bar{y}_5) = -3.15$	For New Effects: $\left(\frac{2}{\sqrt{n}}\right)S =$
$AB = \frac{1}{2}(\bar{y}_3 - \bar{y}_4 + \bar{y}_2 - \bar{y}_5) = -0.85$	For CIM: $\left(\frac{1.78}{\sqrt{n}}\right)S =$
$CIM = \frac{1}{5}(\bar{y}_3 + \bar{y}_4 + \bar{y}_2 + \bar{y}_5 - 4\bar{y}_1) = -0.22$	

Cycle: n=2 Phase 1

Calculation of Averages							Calculation of Standard Deviation
Operating Conditions		(1)	(2)	(3)	(4)	(5)	
(i)	Previous Cycle Sum	60.7	59.8	60.2	64.2	57.5	Previous Sum S=
(ii)	Previous Cycle Average	60.7	59.8	60.2	64.2	57.5	Previous Average =
(iii)	New Observation	59.1	62.8	62.5	64.6	58.3	New S=Range x $f_{k,n}$=1.38
(iv)	Differences	1.6	-3.0	-2.3	-0.4	-0.8	Range=4.6
(v)	New Sums	119.8	122.6	122.7	128.8	115.8	New Sum S=1.38
(vi)	New Averages	59.90	61.30	61.35	64.40	57.90	New average S = New Sum S/(n-1)=1.38

Calculation of Effects	Calculation of Error Limits
$A = \frac{1}{2}(\bar{y}_3 + \bar{y}_4 - \bar{y}_2 - \bar{y}_5) = 3.28$	For New Average: $\left(\frac{2}{\sqrt{n}}\right)S = 1.95$
$B = \frac{1}{2}(\bar{y}_3 - \bar{y}_4 - \bar{y}_2 + \bar{y}_5) = -3.23$	For New Effects: $\left(\frac{2}{\sqrt{n}}\right)S = 1.95$

For CIM: $\left(\frac{1.78}{\sqrt{n}}\right)S = 1.74$

$$CIM = \frac{1}{5}\left(\bar{y}_3 + \bar{y}_4 + \bar{y}_2 + \bar{y}_5 - 4\bar{y}_1\right) = 1.07$$

Cycle: n=3 Phase 1

Calculation of Averages							Calculation of Standard Deviation
Operating Conditions		(1)	(2)	(3)	(4)	(5)	
(i)	Previous Cycle Sum	119.8	122.6	122.7	128.8	115.8	Previous Sum S=1.38
(ii)	Previous Cycle Average	59.90	61.30	61.35	64.40	57.90	Previous Average =1.38
(iii)	New Observation	56.6	59.1	59.0	62.3	61.1	New S=Range x $f_{k,n}$=2.28
(iv)	Differences	3.30	2.20	2.35	2.10	-3.20	Range=6.5
(v)	New Sums	176.4	181.7	181.7	191.1	176.9	New Sum S=3.66
(vi)	New Averages	58.80	60.57	60.57	63.70	58.97	New average S = New Sum S/(n-1)=1.83

Calculation of Effects		Calculation of Error Limits	
$A = \frac{1}{2}\left(\bar{y}_3 + \bar{y}_4 - \bar{y}_2 - \bar{y}_5\right) =$	2.37	For New Average: $\left(\frac{2}{\sqrt{n}}\right)S =$	2.11
$B = \frac{1}{2}\left(\bar{y}_3 - \bar{y}_4 - \bar{y}_2 + \bar{y}_5\right) =$	-2.37	For New Effects: $\left(\frac{2}{\sqrt{n}}\right)S =$	2.11
$AB = \frac{1}{2}\left(\bar{y}_3 - \bar{y}_4 + \bar{y}_2 - \bar{y}_5\right) =$	-0.77	For CIM: $\left(\frac{1.78}{\sqrt{n}}\right)S =$	1.88
$CIM = \frac{1}{5}\left(\bar{y}_3 + \bar{y}_4 + \bar{y}_2 + \bar{y}_5 - 4\bar{y}_1\right) =$	1.72		

Cycle: n=4 Phase 1

Calculation of Averages							Calculation of Standard Deviation
Operating Conditions		(1)	(2)	(3)	(4)	(5)	
(i)	Previous Cycle Sum	176.4	181.7	181.7	191.1	176.9	Previous Sum S=3.66
(ii)	Previous Cycle Average	58.80	60.57	60.57	63.70	58.97	Previous Average =1.83
(iii)	New Observation	60.5	59.8	64.5	61.0	60.1	New S=Range x $f_{k,n}$=2.45
(iv)	Differences	-1.70	0.77	-3.93	2.70	-1.13	Range=6.63
(v)	New Sums	236.9	241.5	246.2	252.1	237.0	New Sum S=6.11
(vi)	New Averages	59.23	60.38	61.55	63.03	59.25	New average S = New Sum S/(n-1)=2.04

Calculation of Effects		Calculation of Error Limits	
$A = \frac{1}{2}\left(\bar{y}_3 + \bar{y}_4 - \bar{y}_2 - \bar{y}_5\right) =$	2.48	For New Average: $\left(\frac{2}{\sqrt{n}}\right)S =$	2.04
$B = \frac{1}{2}\left(\bar{y}_3 - \bar{y}_4 - \bar{y}_2 + \bar{y}_5\right) =$	-1.31	For New Effects: $\left(\frac{2}{\sqrt{n}}\right)S =$	2.04

$$AB = \frac{1}{2}\left(\bar{y}_3 - \bar{y}_4 + \bar{y}_2 - \bar{y}_5\right) = \quad -0.18 \qquad \text{For CIM: } \left(\frac{1.78}{\sqrt{n}}\right)S = \quad 1.82$$

$$CIM = \frac{1}{5}\left(\bar{y}_3 + \bar{y}_4 + \bar{y}_2 + \bar{y}_5 - 4\bar{y}_1\right) = \quad 1.46$$

From studying cycles 3 and 4, it is apparent that *A* (and possibly *B*) has a significant effect. A new phase should be started following cycle 3 or 4.

11.18. Suppose that we approximate a response surface with a model of order d_1, such as $\mathbf{y}=\mathbf{X}_1\boldsymbol{\beta}_1+\boldsymbol{\varepsilon}$, when the true surface is described by a model of order $d_2>d_1$; that is $E(\mathbf{y}) = \mathbf{X}_1\boldsymbol{\beta}_{1+}\ \mathbf{X}_2\boldsymbol{\beta}_2$.

(a) Show that the regression coefficients are biased, that is, that $E(\hat{\boldsymbol{\beta}}_1)=\boldsymbol{\beta}_1+\mathbf{A}\boldsymbol{\beta}_2$, where $\mathbf{A} = (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2$. **A** is usually called the alias matrix.

$$\begin{aligned}
E\left[\hat{\boldsymbol{\beta}}_1\right] &= E\left[\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\mathbf{y}\right] \\
&= \left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1' E\left[\mathbf{y}\right] \\
&= \left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\left(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2\right) \\
&= \left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\mathbf{X}_1\boldsymbol{\beta}_1 + \left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 \\
&= \boldsymbol{\beta}_1 + \mathbf{A}\boldsymbol{\beta}_2
\end{aligned}$$

where $\mathbf{A} = \left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\mathbf{X}_2$

(b) If d_1=1 and d_2=2, and a full 2^k is used to fit the model, use the result in part (a) to determine the alias structure.

In this situation, we have assumed the true surface to be first order, when it is really second order. If a full factorial is used for k=2, then

$$\text{Then, } \mathbf{E}\left[\begin{matrix}\hat{\beta}_0 \\ \beta_1 \\ \hat{\beta}_2\end{matrix}\right] = \mathbf{E}\left[\begin{matrix}\beta_0 \\ \beta_1 \\ \beta_2\end{matrix}\right] + \left[\begin{matrix}1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0\end{matrix}\right]\left[\begin{matrix}\beta_{11} \\ \beta_{22} \\ \beta_{12}\end{matrix}\right] = \left[\begin{matrix}\beta_0 + \beta_{11} + \beta_{22} \\ \beta_1 \\ \beta_2\end{matrix}\right]$$

The pure quadratic terms bias the intercept.

(c) If d_1=1, d_2=2 and k=3, find the alias structure assuming that a 2^{3-1} design is used to fit the model.

$$\mathbf{X}_1 = \begin{matrix} \beta_0 & \beta_1 & \beta_2 & \beta_3 \end{matrix} \begin{bmatrix} 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \mathbf{X}_2 = \begin{matrix} \beta_{11} & \beta_{22} & \beta_{33} & \beta_{12} & \beta_{13} & \beta_{23} \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\text{and } \mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\text{Then, } \left[\hat{\beta}_1\right] = \mathbf{E}\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \beta_{11} \\ \beta_{22} \\ \beta_{33} \\ \beta_{12} \\ \beta_{13} \\ \beta_{23} \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_{11} + \beta_{22} + \beta_{33} \\ \beta_1 + \beta_{23} \\ \beta_2 + \beta_{13} \\ \beta_3 + \beta_{12} \end{bmatrix}$$

(d) If d_1=1, d_2=2, k=3, and the simplex design in Problem 11.3 is used to fit the model, determine the alias structure and compare the results with part (c).

$$\mathbf{X}_1 = \begin{matrix} \beta_0 & \beta_1 & \beta_2 & \beta_3 \end{matrix} \begin{bmatrix} 0 & 0 & \sqrt{2} & -1 \\ 2 & -\sqrt{2} & 0 & 1 \\ 0 & 0 & -\sqrt{2} & -1 \\ 2 & -\sqrt{2} & 0 & 1 \end{bmatrix} \quad \mathbf{X}_2 = \begin{matrix} \beta_{11} & \beta_{22} & \beta_{33} & \beta_{12} & \beta_{13} & \beta_{23} \end{matrix} \begin{bmatrix} 0 & 2 & 1 & 0 & 0 & -\sqrt{2} \\ 2 & 0 & 1 & 0 & -\sqrt{2} & 0 \\ 0 & 2 & 1 & 0 & 0 & -\sqrt{2} \\ 2 & 0 & 1 & 0 & -\sqrt{2} & 0 \end{bmatrix}$$

$$\text{and } \mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & -1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Then, } \mathbf{E}\left[\hat{\beta}_1\right] = \mathbf{E}\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & -1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \beta_{11} \\ \beta_{22} \\ \beta_{33} \\ \beta_{12} \\ \beta_{13} \\ \beta_{23} \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_{11} + \beta_{22} + \beta_{33} \\ \beta_1 + \beta_{13} \\ \beta_2 - \beta_{23} \\ \beta_3 + \beta_{11} - \beta_{22} \end{bmatrix}$$

Notice that the alias structure is different from that found in the previous part for the 2^{3-1} design. In general, the **A** matrix will depend on which simplex design is used.

11.19S. Suppose that you need to design an experiment to fit a quadratic model over the region $-1 \le x_i \le +1$, i = 1, 2 subject to the constraint $x_1 + x_2 \le 1$. If the constraint is violated, the process will not work properly. You can afford to make no more than n = 12 runs. Set up the following designs:

(a) An "inscribed" CCD with center points at $x_1 = x_2 = 0$

x_1	x_2
-0.5	-0.5

0.5	-0.5
-0.5	0.5
0.5	0.5
-0.707	0
0.707	0
0	-0.707
0	0.707
0	0
0	0
0	0
0	0

(a)* An "inscribed" CCD with center points at $x_1 = x_2 = -0.25$ so that a larger design could be fit within the constrained region

x_1	x_2
-1	-1
0.5	-1
-1	0.5
0.5	0.5
-1.664	-0.25
1.164	-0.25
-0.25	-1.664
-0.25	1.164
-0.25	-0.25
-0.25	-0.25
-0.25	-0.25
-0.25	-0.25

(b) An "inscribed" 3^2 factorial with center points at $x_1 = x_2 - 0.25$

x_1	x_2
-1	-1
-0.25	-1
0.5	-1
-1	-0.25
-0.25	-0.25
0.5	-0.25
-1	0.5
-0.25	0.5
0.5	0.5
-0.25	-0.25
-0.25	-0.25
-0.25	-0.25

(c) A *D*-optimal design.

x_1	x_2
-1	-1
1	-1
-1	1
1	0
0	1
0	0
-1	0
0	-1
0.5	0.5

-1	-1
1	-1
-1	1

(d) A modified *D*-optimal design that is identical to the one in part (c), but with all replicate runs at the design center.

x_1	x_2
1	0
0	0
0	1
-1	-1
1	-1
-1	1
-1	0
0	-1
0.5	0.5
0	0
0	0
0	0

(e) Evaluate the $\left|(\mathbf{X}'\mathbf{X})^{-1}\right|$ criteria for each design.

	(a)	(a)*	(b)	(c)	(d)
$\left\|(\mathbf{X}'\mathbf{X})^{-1}\right\|$	0.5	0.00005248	0.007217	0.0001016	0.0002294

(f) Evaluate the *D*-efficiency for each design relative to the *D*-optimal design in part (c).

	(a)	(a)*	(b)	(c)	(d)
D-efficiency	24.25%	111.64%	49.14%	100.00%	87.31%

(g) Which design would you prefer? Why?

The offset CCD, (a)*, is the preferred design based on the *D*-efficiency. Not only is it better than the *D*-optimal design, (c), but it maintains the desirable design features of the CCD.

11.20S. Consider a 2^3 design for fitting a first-order model.

(a) Evaluate the *D*-criterion $\left|(\mathbf{X}'\mathbf{X})^{-1}\right|$ for this design.

$$\left|(\mathbf{X}'\mathbf{X})^{-1}\right| = 2.441E-4$$

(b) Evaluate the *A*-criterion $tr(\mathbf{X}'\mathbf{X})^{-1}$ for this design.

$$tr(\mathbf{X}'\mathbf{X})^{-1} = 0.5$$

(c) Find the maximum scaled prediction variance for this design. Is this design *G*-optimal?

$$v(\mathbf{x}) = \frac{NVar\left(\hat{y}(\mathbf{x})\right)}{\sigma^2} = N\mathbf{x}'^{(1)}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{x}^{(1)} = 4.$$

Yes, this is a *G*-optimal design.

11.21. Repeat Problem 11.20 using a first order model with the two-factor interaction.

$$\left|(\mathbf{X}'\mathbf{X})^{-1}\right| = 4.768E-7$$

$$tr(\mathbf{X}'\mathbf{X})^{-1} = 0.875$$

$$v(\mathbf{x}) = \frac{NVar\left(\hat{y}(\mathbf{x})\right)}{\sigma^2} = N\mathbf{x}'^{(1)}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{x}^{(1)} = 7.$$

Yes, this is a *G*-optimal design.

11.22S. Suppose that you want to fit a *second*-order model in $k = 5$ factors. You cannot afford more than 25 runs. Construct both a *D*-optimal and an *I*-optimal design for this situation. Compare the prediction variance properties of the designs. Which design would you prefer?

The following JMP outputs identify both *D*-optimal and *I*-optimal designs. The prediction variance profile shown in each JMP output is appreciably less for the *I*-optimal design as well as flatter in the middle of the design.

JMP Output

D-optimal Design

Factors

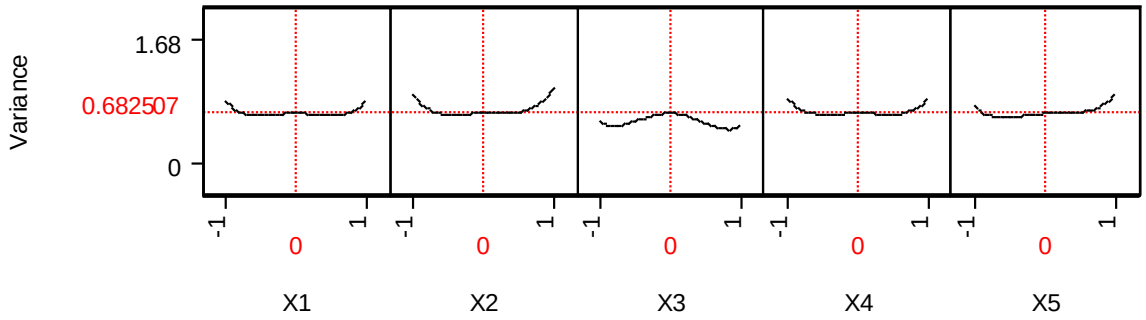
Add N Factors
1

X1 Continuous
X2 Continuous
X3 Continuous
X4 Continuous
X5 Continuous

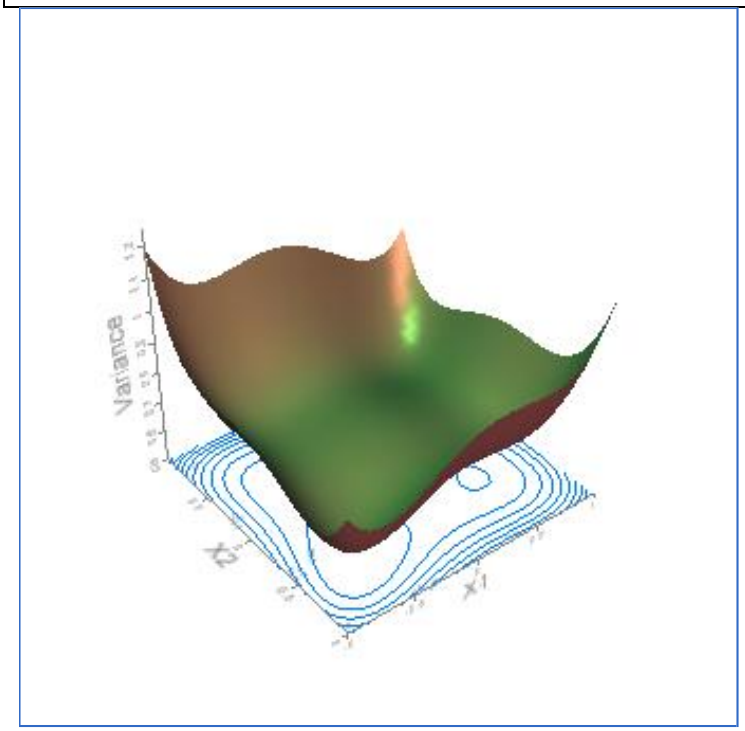
Design

Run	X1	X2	X3	X4	X5
1	-1	-1	1	1	0
2	-1	1	1	-1	-1
3	-1	1	-1	-1	1
4	1	-1	-1	-1	1
5	-1	-1	-1	1	1
6	0	1	1	-1	0
7	0	-1	0	1	-1
8	-1	-1	-1	-1	-1
9	1	-1	-1	1	-1
10	1	-1	1	1	1
11	-1	-1	1	-1	1
12	1	-1	1	-1	-1
13	-1	1	-1	1	-1
14	1	1	1	1	-1
15	0	-1	-1	0	0
16	-1	0	0	-1	0
17	1	1	1	-1	1
18	1	1	0	0	-1
19	-1	-1	1	0	-1
20	-1	1	1	1	1
21	1	1	-1	1	1
22	1	1	-1	-1	-1
23	0	0	1	0	1
24	0	0	1	1	-1
25	1	0	-1	0	0

Prediction Variance Profile



Prediction Variance Surface



JMP Output

I-Optimal Design

Factors

Add N Factors
1

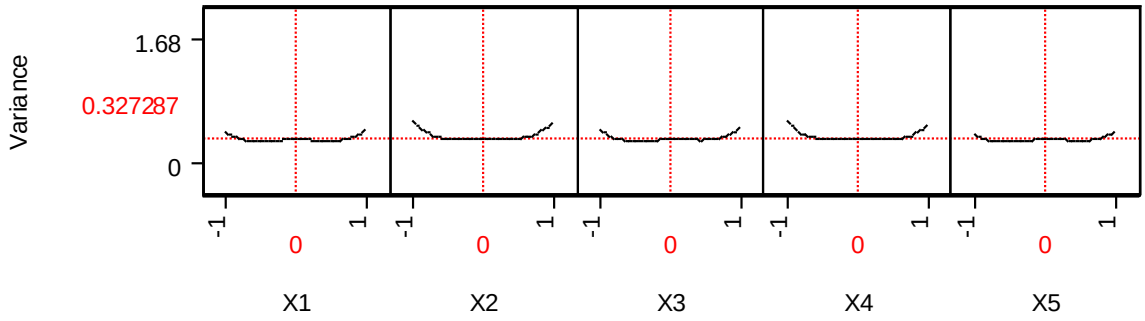
X1 Continuous
X2 Continuous
X3 Continuous
X4 Continuous
X5 Continuous

Design

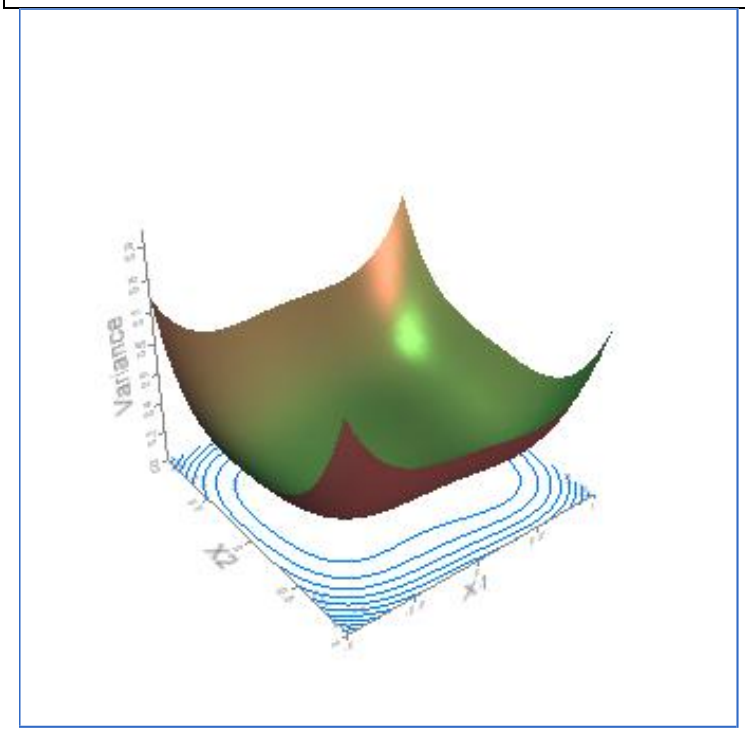
Run	X1	X2	X3	X4	X5
1	-1	0	-1	1	0
2	1	1	1	-1	-1
3	1	1	-1	1	-1
4	0	-1	0	-1	0
5	-1	1	0	0	0
6	0	0	0	1	1

7	-1	0	1	-1	1
8	1	-1	-1	1	1
9	1	0	0	0	-1
10	1	0	-1	-1	0
11	-1	-1	-1	-1	1
12	-1	-1	1	-1	-1
13	1	-1	1	0	1
14	1	1	0	-1	1
15	0	1	1	1	-1
16	-1	0	0	0	-1
17	1	1	1	1	0
18	0	1	-1	0	1
19	1	-1	1	1	-1
20	0	0	0	0	0
21	-1	1	1	1	1
22	-1	1	-1	-1	-1
23	-1	-1	1	1	0
24	0	-1	-1	0	-1
25	0	0	1	0	0

Prediction Variance Profile



Prediction Variance Surface



11.23. Suppose that you want to fit a *second*-order response surface model in a situation where there are k = 4 factors; however, one of the factors is categorical with two levels. What model should you consider for this experiment? Suggest an appropriate design for this situation.

The following model is considered for this experiment:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \gamma z + \delta_1 x_1 z + \delta_2 x_2 z + \delta_3 x_3 z + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{33} x_3^2 + \varepsilon u$$

The following JMP output identifies a computer generated design utilizing the *I*-criterion. Because this is a response surface model which is generally used for prediction, the *I*-criterion is an appropriate choice for a computer generated design.

JMP Output

Custom Design

Factors

Add N Factors
1

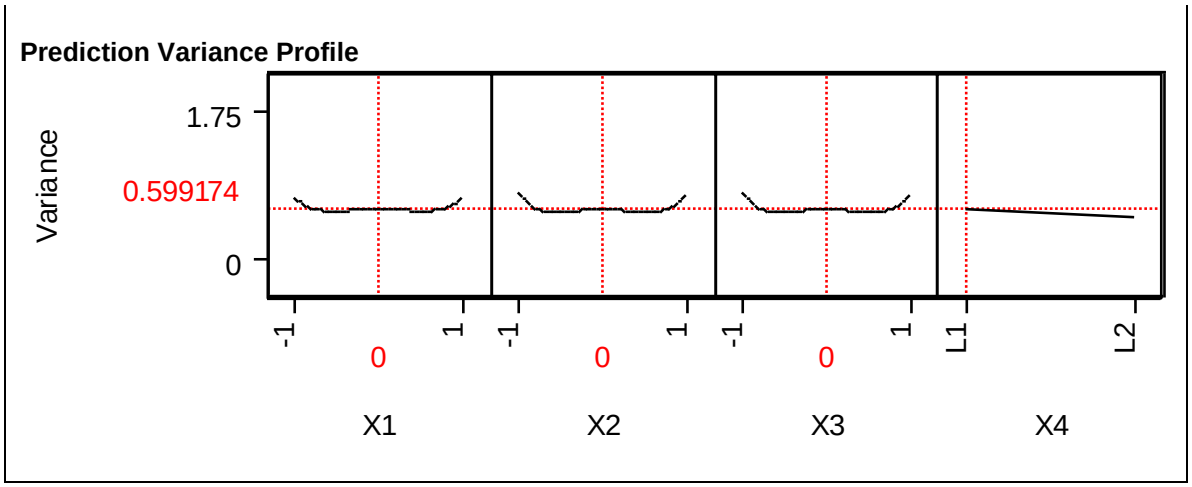
X1 Continuous
X2 Continuous
X3 Continuous
X4 Categorical

Model

Intercept
Intercept
X1
X2
X3
X4
X1*X2
X1*X3
X1*X4
X2*X3
X2*X4
X3*X4
X1*X1
X2*X2
X3*X3

Design

Run	X1	X2	X3	X4
1	0	0	0	L2
2	-1	-1	-1	L2
3	1	-1	1	L1
4	1	1	1	L2
5	0	-1	-1	L1
6	0	1	1	L1
7	-1	0	1	L1
8	0	1	-1	L2
9	-1	-1	0	L1
10	0	-1	1	L2
11	-1	1	-1	L1
12	1	-1	-1	L2
13	1	0	-1	L1
14	0	0	0	L2
15	1	1	0	L1
16	-1	1	1	L2



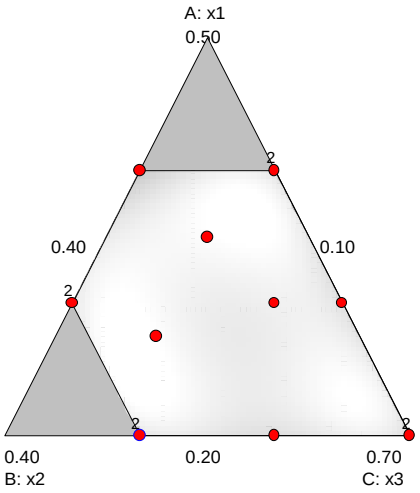
11.24S. An experimenter wishes to run a three-component mixture experiment. The constraints in the components proportions are as follows:

$$0.2 \le x_1 \le 0.4$$
$$0.1 \le x_2 \le 0.3$$
$$0.4 \le x_3 \le 0.7$$

(a) Set up an experiment to fit a quadratic mixture model. Use n = 14 runs, with 4 replicates. Use the D-criteria.

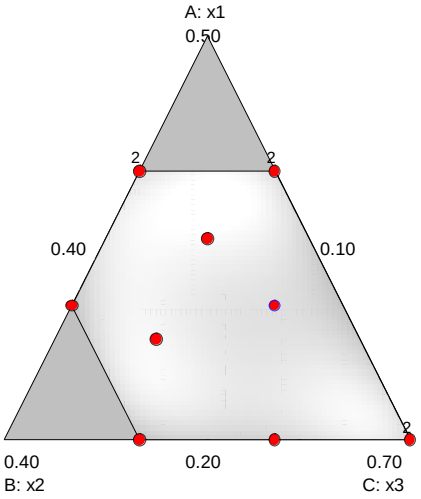
Std	x1	x2	x3
1	0.2	0.3	0.5
2	0.3	0.3	0.4
3	0.3	0.15	0.55
4	0.2	0.1	0.7
5	0.4	0.2	0.4
6	0.4	0.1	0.5
7	0.2	0.2	0.6
8	0.275	0.25	0.475
9	0.35	0.175	0.475
10	0.3	0.1	0.6
11	0.2	0.3	0.5
12	0.3	0.3	0.4
13	0.2	0.1	0.7
14	0.4	0.1	0.5

(b) Draw the experimental design region.



(c) Set up an experiment to fit a quadratic mixture model with $n = 12$ runs, assuming that three of these runs are replicated. Use the *D*-criterion.

Std	x_1	x_2	x_3
1	0.3	0.15	0.55
2	0.2	0.3	0.5
3	0.3	0.3	0.4
4	0.2	0.1	0.7
5	0.4	0.2	0.4
6	0.4	0.1	0.5
7	0.2	0.2	0.6
8	0.275	0.25	0.475
9	0.35	0.175	0.475
10	0.2	0.1	0.7
11	0.4	0.1	0.5
12	0.4	0.2	0.4



(d) Comment on the two designs you have found.

The design points are the same for both designs except that the edge center on the x1-x3 edge is not included in the second design. None of the replicates for either design are in the center of the experimental region. The experimental runs are fairly uniformly spaced in the design region.

11.25. Myers, Montgomery and Anderson-Cook (2009) describe a gasoline blending experiment involving three mixture components. There are no constraints on the mixture proportions, and the following 10 run design is used.

Design Point	x_1	x_2	x_3	y(mpg)
1	1	0	0	24.5, 25.1
2	0	1	0	24.8, 23.9
3	0	0	1	22.7, 23.6
4	½	½	0	25.1
5	½	0	½	24.3
6	0	½	½	23.5
7	1/3	1/3	1/3	24.8, 24.1
8	2/3	1/6	1/6	24.2
9	1/6	2/3	1/6	23.9
10	1/6	1/6	2/3	23.7

(a) What type of design did the experimenters use?

A simplex centroid design was used.

(b) Fit a quadratic mixture model to the data. Is this model adequate?

Design Expert Output

Response: y

ANOVA for Mixture Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	4.22	5	0.84	3.90	0.0435	significant
Linear Mixture	*3.92*	*2*	*1.96*	*9.06*	*0.0088*	
AB	*0.15*	*1*	*0.15*	*0.69*	*0.4289*	

AC	*0.081*	*1*	*0.081*	*0.38*	*0.5569*	
BC	*0.077*	*1*	*0.077*	*0.36*	*0.5664*	
Residual	1.73	8	0.22			
Lack of Fit	*0.50*	*4*	*0.12*	*0.40*	*0.8003*	*not significant*
Pure Error	*1.24*	*4*	*0.31*			
Cor Total	5.95	13				

The Model F-value of 3.90 implies the model is significant. There is only a 4.35% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.47	R-Squared	0.7091
Mean	24.16	Adj R-Squared	0.5274
C.V.	1.93	Pred R-Squared	0.1144
PRESS	5.27	Adeq Precision	5.674

Component	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High
A-x1	24.74	1	0.32	24.00	25.49
B-x2	24.31	1	0.32	23.57	25.05
C-x3	23.18	1	0.32	22.43	23.92
AB	1.51	1	1.82	-2.68	5.70
AC	1.11	1	1.82	-3.08	5.30
BC	-1.09	1	1.82	-5.28	3.10

Final Equation in Terms of Pseudo Components:

```
      y =
+24.74  * A
+24.31  * B
+23.18  * C
 +1.51  * A * B
 +1.11  * A * C
 -1.09  * B * C
```

Final Equation in Terms of Real Components:

```
       y =
+24.74432  * x1
+24.31098  * x2
+23.17765  * x3
 +1.51364  * x1 * x2
 +1.11364  * x1 * x3
 -1.08636  * x2 * x3
```

The quadratic terms appear to be insignificant. The analysis below is for the linear mixture model:

Design Expert Output

Response: y

ANOVA for Mixture Quadratic Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Model	3.92	2	1.96	10.64	0.0027	significant
Linear Mixture	*3.92*	*2*	*1.96*	*10.64*	*0.0027*	
Residual	2.03	11	0.18			
Lack of Fit	*0.79*	*7*	*0.11*	*0.37*	*0.8825*	*not significant*
Pure Error	*1.24*	*4*	*0.31*			
Cor Total	5.95	13				

The Model F-value of 10.64 implies the model is significant. There is only a 0.27% chance that a "Model F-Value" this large could occur due to noise.

Std. Dev.	0.43	R-Squared	0.6591
Mean	24.16	Adj R-Squared	0.5972
C.V.	1.78	Pred R-Squared	0.3926
PRESS	3.62	Adeq Precision	8.751

Component	Coefficient Estimate	DF	Standard Error	95% CI Low	95% CI High
A-x1	24.93	1	0.25	24.38	25.48
B-x2	24.35	1	0.25	23.80	24.90
C-x3	23.19	1	0.25	22.64	23.74

Component	Effect	DF	Adjusted Std Error	Adjusted Effect=0	Approx t for H0 Prob > |t|
A-x1	1.16	1	0.33	3.49	0.0051
B-x2	0.29	1	0.33	0.87	0.4021
C-x3	-1.45	1	0.33	-4.36	0.0011

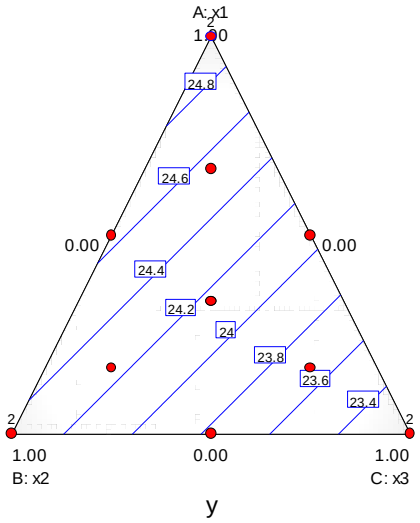
Final Equation in Terms of Pseudo Components:

y =
+24.93 * A
+24.35 * B
+23.19 * C

Final Equation in Terms of Real Components:

y =
+24.93048 * x1
+24.35048 * x2
+23.19048 * x3

(c) Plot the response surface contours. What blend would you recommend to maximize the MPG?



To maximize the miles per gallon, the recommended blend is $x_1 = 1$, $x_2 = 0$, and $x_3 = 0$.

11.26. Table P11.9 shows a six-variable RSM design from Jones and Nachtsheim (2011b). Analyze the response data from this experiment.

Table P11.9 – The Design for Problem 11.26

Run(i)	$x_{i,1}$	$x_{i,2}$	$x_{i,3}$	$x_{i,4}$	$x_{i,5}$	$x_{i,6}$	y_i
1	0	1	-1	-1	-1	-1	21.04
2	0	-1	1	1	1	1	10.48
3	1	0	-1	1	1	-1	17.89
4	-1	0	1	-1	-1	1	10.07
5	-1	-1	0	1	-1	-1	7.74

6	1	1	0	-1	1	1	21.01
7	-1	1	1	0	1	-1	16.53
8	1	-1	-1	0	-1	1	20.38
9	1	-1	1	-1	0	-1	8.62
10	-1	1	-1	1	0	1	7.8
11	1	1	1	1	-1	0	23.56
12	-1	-1	-1	-1	1	0	15.24
13	0	0	0	0	0	0	19.91

Design Expert Output

Response 1 Yi

ANOVA for Response Surface Reduced 2FI Model

Analysis of variance table [Partial sum of squares - Type III]

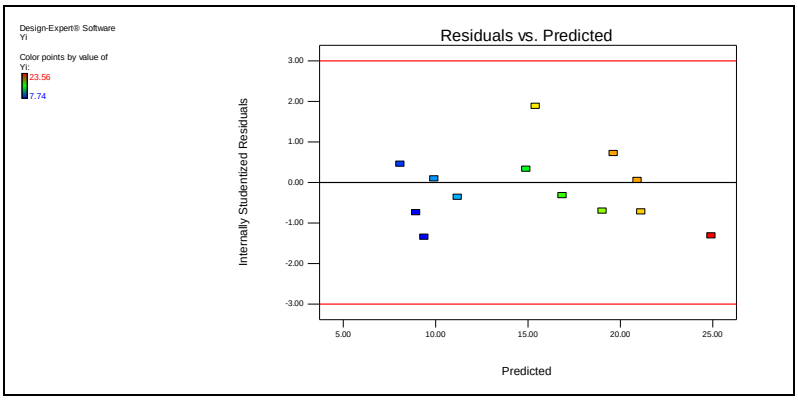
Source	**Sum of Squares**	**df**	**Mean Square**	**F Value**	**p-value Prob > F**	
Model	366.59	7	52.37	8.46	0.0159	significant
A-Xi.1	*116.14*	*1*	*116.14*	*18.77*	*0.0075*	
B-Xi.2	*75.52*	*1*	*75.52*	*12.20*	*0.0174*	
C-Xi.3	*17.13*	*1*	*17.13*	*2.77*	*0.1570*	
E-Xi.5	*0.27*	*1*	*0.27*	*0.043*	*0.8431*	
AB	*84.75*	*1*	*84.75*	*13.70*	*0.0140*	
AC	*64.42*	*1*	*64.42*	*10.41*	*0.0233*	
AE	*109.00*	*1*	*109.00*	*17.61*	*0.0085*	
Residual	30.94	5	6.19			
Cor Total	397.53	12				

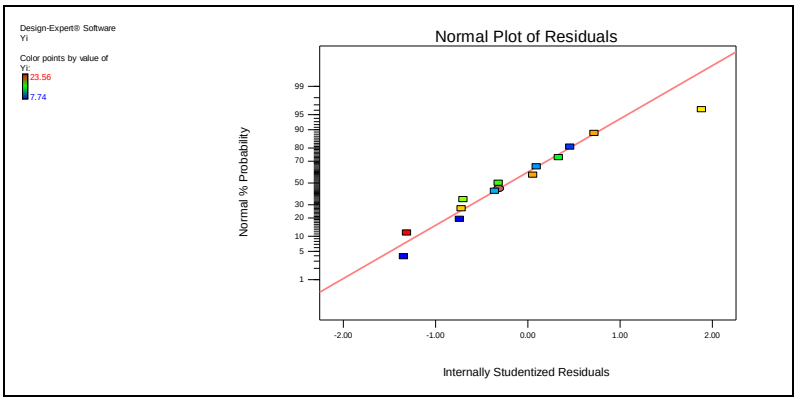
Std. Dev.	2.49	R-Squared	0.9222
Mean	15.41	Adj R-Squared	0.8132
C.V. %	16.15	Pred R-Squared	0.5242
PRESS	189.14	Adeq Precision	8.627

Factor	**Coefficient Estimate**	**df**	**Error**	**Standard Low**	**95% CI High**	**95% CI VIF**
Intercept	15.41	1	0.69	13.63	17.18	
A-Xi.1	3.41	1	0.79	1.39	5.43	1.00
B-Xi.2	2.75	1	0.79	0.73	4.77	1.00
C-Xi.3	-1.31	1	0.79	-3.33	0.71	1.00
E-Xi.5	-0.16	1	0.79	-2.19	1.86	1.00
AB	3.57	1	0.96	1.09	6.04	1.20
AC	-3.11	1	0.96	-5.59	-0.63	1.20
AE	-4.04	1	0.96	-6.52	-1.57	1.20

Final Equation in Terms of Actual Factors:

Yi =
+15.40538
+3.40800 * Xi.1
+2.74800 * Xi.2
-1.30900 * Xi.3
-0.16400 * Xi.5
+3.56550 * Xi.1 * Xi.2
-3.10850 * Xi.1 * Xi.3
-4.04350 * Xi.1 * Xi.5





The main effects *A* and *B* are significant as well as the *AB*, *AC* and *AE* two-factor interactions. Factors *C* and *E* are kept in the model to maintain hierarchy. The normal probability plot of residuals shows a possible outlier in the upper right corner.

11.27. An article in *Quality Progress* ("For Starbucks, It's in the Bag," March 2011, pp. 18-23) describes using a central composite design to improve the packaging of one-pound coffee. The objective is to produce an airtight seal that is easy to open without damaging the top of the coffee bag. The experimenters studied the three factors – x_1 = plastic viscosity (300-400 centipoise), x_2 = clamp pressure (170-190 psi), and x_3 = plate gap (-3, +3 mm) and two responses – y_1 = tear and y_2 = leakage. The design is shown in Table P11.10. The tear response was measured on a scale from 0 to 9 (good to bad) and the leakage was proportion failing. Each run used a sample of 20 bags for response measurement.

Table P11.10 – The Coffee Bag Experiment in Problem 11.27

Run	Viscosity	Pressure	Plate gap	Tear	Leakage
Center	350	180	0	0	0.15
Axial	350	170	0	0	0.5
Factorial	319	186	1.8	0.45	0.15
Factorial	380	174	1.8	0.85	0.05
Center	350	180	0	0.35	0.15
Axial	300	180	0	0.3	0.45
Axial	400	180	0	0.7	0.25
Axial	350	190	0	1.9	0
Center	350	180	0	0.25	0.05
Factorial	319	186	-1.8	0.1	0.35
Factorial	380	186	-1.8	0.15	0.4
Axial	350	180	3	3.9	0
Factorial	380	174	-1.8	0	0.45
Center	350	180	0	0.55	0.2

Axial	350	180	-3	0	1
Factorial	319	174	-1.8	0.05	0.2
Factorial	319	174	1.8	0.4	0.25
Factorial	380	186	1.8	4.3	0.05
Center	350	180	0	0	0

(a) Build a second-order model for the tear response.

Design Expert Output

Response 1 Tear

ANOVA for Response Surface Reduced Cubic Model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	26.40	8	3.30	13.72	0.0002	significant
A-Viscosity	*1.81*	*1*	*1.81*	*7.52*	*0.0207*	
B-Pressure	*3.48*	*1*	*3.48*	*14.47*	*0.0035*	
C-Plate Gap	*11.00*	*1*	*11.00*	*45.73*	*< 0.0001*	
AB	*1.53*	*1*	*1.53*	*6.36*	*0.0303*	
AC	*2.31*	*1*	*2.31*	*9.60*	*0.0113*	
BC	*1.36*	*1*	*1.36*	*5.66*	*0.0387*	
C2	*3.54*	*1*	*3.54*	*14.72*	*0.0033*	
ABC	*1.36*	*1*	*1.36*	*5.66*	*0.0387*	
Residual	2.41	10	0.24			
Lack of Fit	*2.18*	*6*	*0.36*	*6.53*	*0.0454*	*significant*
Pure Error	*0.22*	*4*	*0.056*			
Cor Total	28.81	18				

Std. Dev.	0.49	R-Squared	0.9165
Mean	0.75	Adj R-Squared	0.8497
C.V. %	65.41	Pred R-Squared	0.4004
PRESS	17.27	Adeq Precision	14.560

Factor	Coefficient Estimate	df	Error	Standard Low	95% CI High	95% CI VIF
Intercept	0.39	1	0.15	0.065	0.72	
A-Viscosity	0.36	1	0.13	0.068	0.66	1.00
B-Pressure	0.50	1	0.13	0.21	0.80	1.00
C-Plate Gap	0.90	1	0.13	0.60	1.19	1.00
AB	0.44	1	0.17	0.051	0.82	1.00
AC	0.54	1	0.17	0.15	0.92	1.00
BC	0.41	1	0.17	0.026	0.80	1.00
C2	0.50	1	0.13	0.21	0.79	1.00
ABC	0.41	1	0.17	0.026	0.80	1.00

Final Equation in Terms of Actual Factors:

Tear =

+131.47081

-0.41839 * Viscosity

-0.75140 * Pressure

+68.98264 * Plate Gap

+2.39071E-003 * Viscosity * Pressure

-0.21562 * Viscosity * Plate Gap

-0.39948 * Pressure * Plate Gap

+0.15424 * Plate Gap^2

+1.25228E-003 * Viscosity * Pressure * Plate Gap

The second-order model for the tear response includes all three main effects, all two-factor interactions, the three-factor interaction and B^2.

(b) Build a second-order model for the leakage response.

Design Expert Output

Response 2 Leakage

ANOVA for Response Surface Reduced Quadratic Model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	0.62	2	0.31	10.94	0.0010	significant
C-Plate Gap	*0.49*	*1*	*0.49*	*17.13*	*0.0008*	
C2	*0.14*	*1*	*0.14*	*4.75*	*0.0445*	
Residual	0.46	16	0.028			
Lack of Fit	*0.43*	*12*	*0.036*	*5.30*	*0.0603*	*not significant*
Pure Error	*0.027*	*4*	*6.750E-003*			
Cor Total	1.08	18				

Std. Dev.	0.17	R-Squared	0.5776
Mean	0.24	Adj R-Squared	0.5248
C.V. %	68.98	Pred R-Squared	0.1949
PRESS	0.87	Adeq Precision	10.222

Factor	Coefficient Estimate	df	Error	Standard Low	95% CI High	95% CI VIF
Intercept	0.17	1	0.050	0.068	0.28	
C-Plate Gap	-0.19	1	0.046	-0.29	-0.092	1.00
C2	0.098	1	0.045	2.704E-003	0.19	1.00

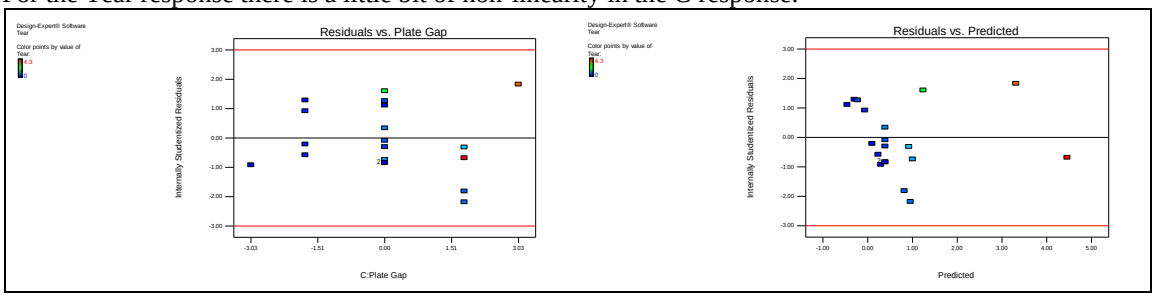
Final Equation in Terms of Actual Factors:

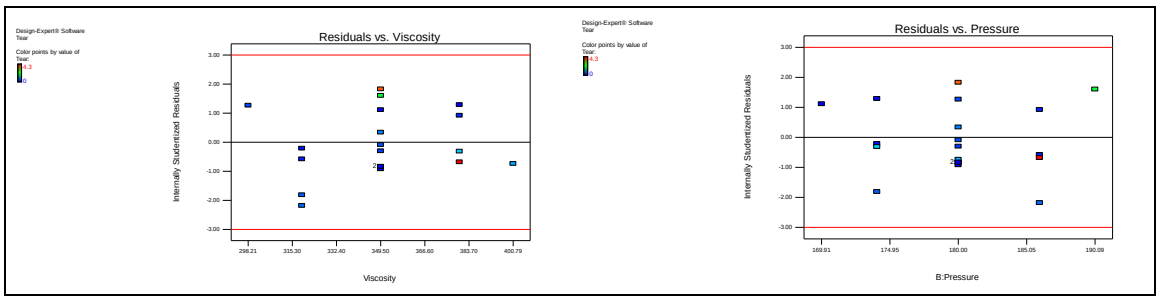
Leakage =
+0.17449
-0.10503 * Plate Gap
+0.030162 * Plate Gap2

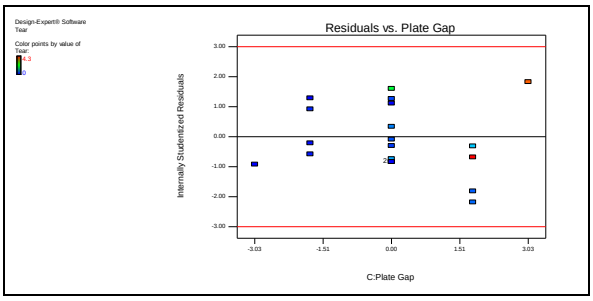
For the Leakage response, only variable C – Plate Gap and C^2 are important.

(c) Analyze the residuals for both models. Do transformations seem necessary for either response? If so, refit the models in the transformed metric.

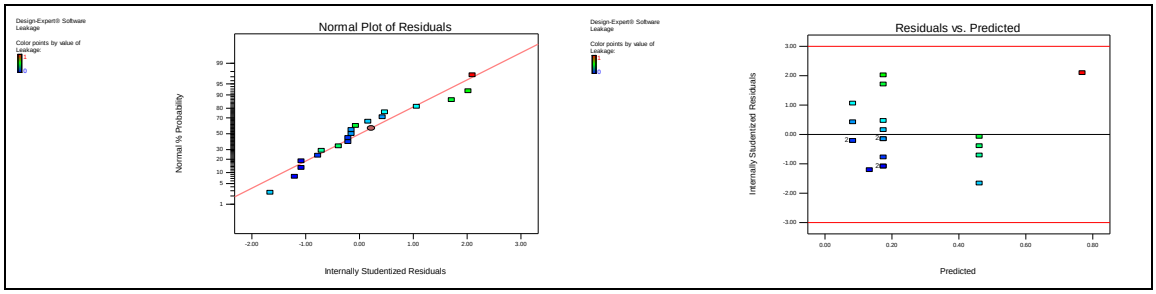
For the Tear response there is a little bit of non-linearity in the C response.

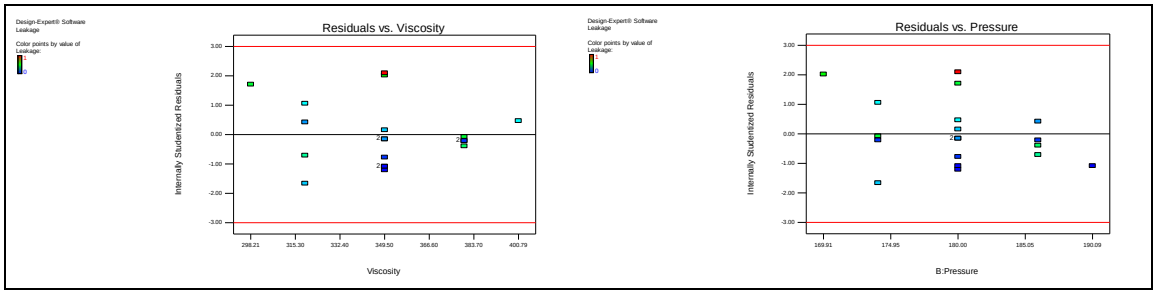


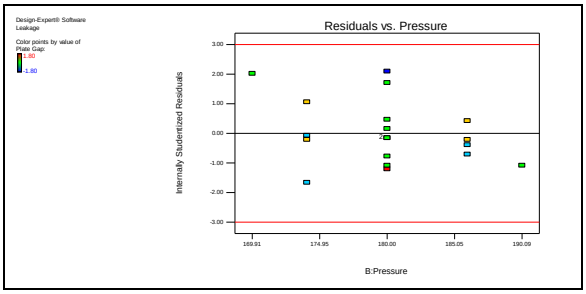




For the Tear response, the residuals look fine. No concerns with model adequacy.



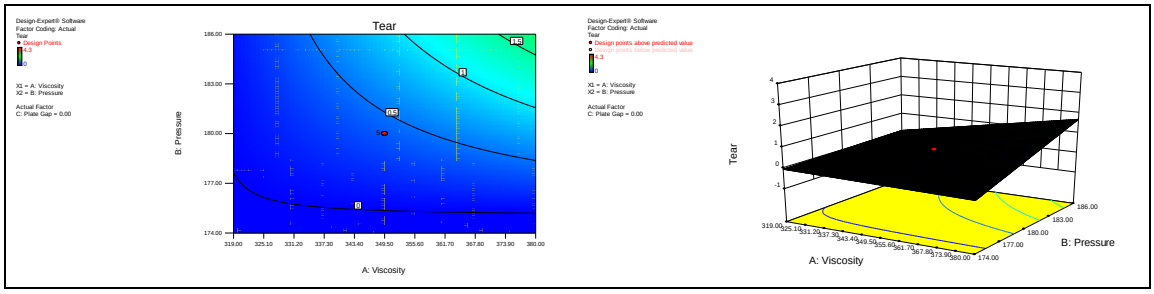




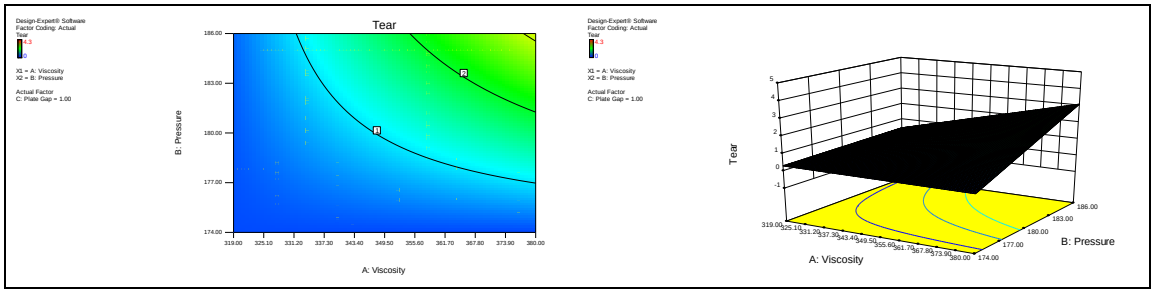
(d) Construct response surface plots and contour plots for both responses. Provide interpretations for the fitted surfaces.

For the Tear Response:

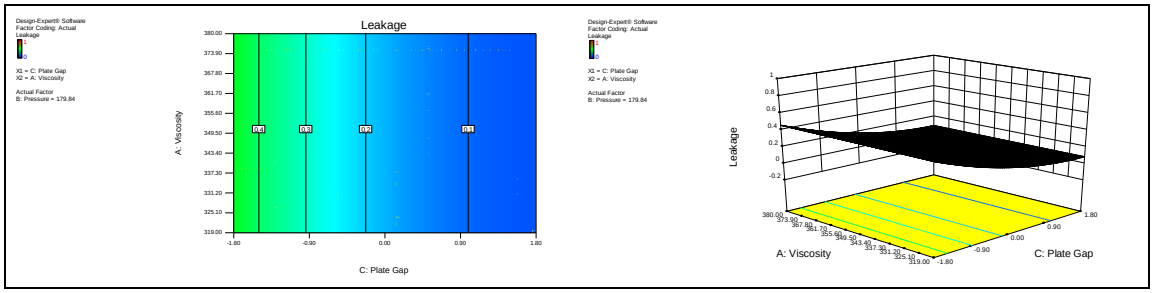
A and *B* with *C*=0



A and *B* with *C*=1

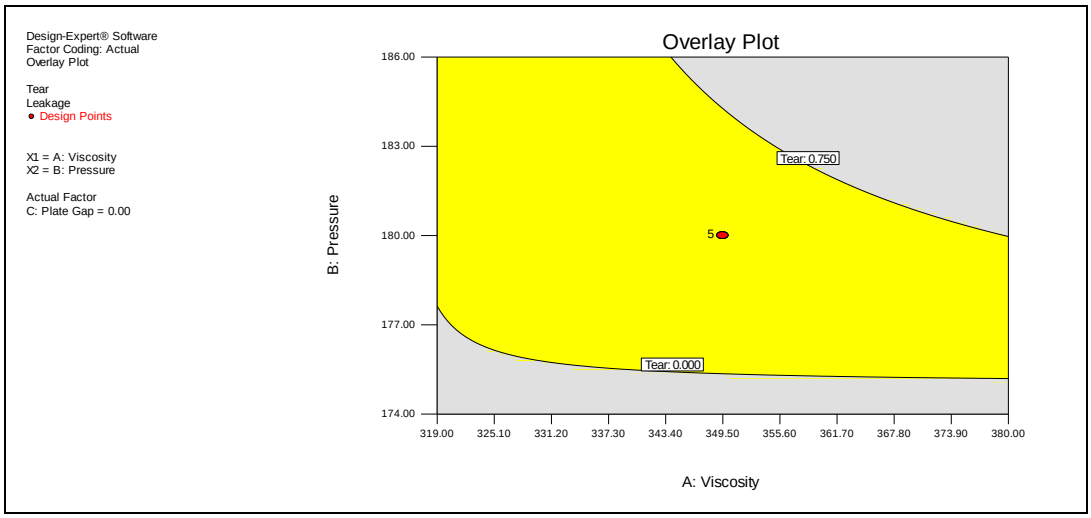


For the Leakage Response, factors *A* and *C*, with *B*=0



(e) What conditions would you recommend for process optimization to minimize leakage and keep tear below 0.75.

Recommend running the process with *C*<0. When *C* is >0 there is very little range where the Tear response is less than 0.75. The graph below shows the *A* and *B* ranges when *C*=0.



11.28. Box and Liu (1999) describe an experiment flying paper helicopters where the objective is to maximize flight time. They used a central composite design shown in Table P11.11. Each run involved a single helicopter made to the following specifications: x_1 = wing area (in^2), -1 = 11.80 and +1 = 13.00; x_2 = wing-length to width ratio, -1 = 2.25 and +1 = 2.78; x_3 = base width (in), -1 = 1.00 and +1 = 1.50; and x_4 = base length (in), -1 = 1.50 and +1 = 2.50. Each helicopter was flown four times and the average flight time and the standard deviation of flight time was recorded.

Table P11.11- The Paper Helicopter Experiment

Std Ord	Run Ord	Wing Area	Wing Ratio	Base Width	Base Length	Avg Flight Time	St Dev Flight Time
1	9	-1	-1	-1	-1	3.67	0.052
2	21	1	-1	-1	-1	3.69	0.052
3	14	-1	1	-1	-1	3.74	0.055
4	4	1	1	-1	-1	3.7	0.062
5	2	-1	-1	1	-1	3.72	0.052
6	19	1	-1	1	-1	3.55	0.065
7	22	-1	1	1	-1	3.97	0.052
8	25	1	1	1	-1	3.77	0.098
9	27	-1	-1	-1	1	3.5	0.079
10	13	1	-1	-1	1	3.73	0.072
11	20	-1	1	-1	1	3.58	0.083
12	6	1	1	-1	1	3.63	0.132
13	12	-1	-1	1	1	3.44	0.058
14	17	1	-1	1	1	3.55	0.049
15	26	-1	1	1	1	3.7	0.081
16	1	1	1	1	1	3.62	0.051
17	8	-2	0	0	0	3.61	0.129
18	15	2	0	0	0	3.64	0.085
19	7	0	-2	0	0	3.55	0.1
20	5	0	2	0	0	3.73	0.063
21	29	0	0	-2	0	3.61	0.051
22	28	0	0	2	0	3.6	0.095
23	16	0	0	0	-2	3.8	0.049
24	18	0	0	0	2	3.6	0.055
25	24	0	0	0	0	3.77	0.032
26	10	0	0	0	0	3.75	0.055
27	23	0	0	0	0	3.7	0.072

28	11	0	0	0	0	3.68	0.055
29	3	0	0	0	0	3.69	0.078
30	30	0	0	0	0	3.66	0.058

(a) Fit a second-order model to the average flight time response.

The flight time response is driven by B and D, several two-factor interactions and A^2 and C^2 terms. A and C are in the model to maintain hierarchy.

Design Expert Output

Response 1 Avg Flt Time

ANOVA for Response Surface Reduced Quadratic Model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	0.28	11	0.025	15.78	< 0.0001	significant
A-Wing Area	*1.667E-005*	*1*	*1.667E-005*	*0.010*	*0.9198*	
B-Length/Width	*0.062*	*1*	*0.062*	*38.83*	*< 0.0001*	
C-Base Width	*1.500E-004*	*1*	*1.500E-004*	*0.094*	*0.7628*	
D-Base Length	*0.089*	*1*	*0.089*	*55.61*	*< 0.0001*	
AB	*0.013*	*1*	*0.013*	*8.28*	*0.0100*	
AC	*0.023*	*1*	*0.023*	*14.09*	*0.0015*	
AD	*0.031*	*1*	*0.031*	*19.17*	*0.0004*	
BC	*0.034*	*1*	*0.034*	*21.43*	*0.0002*	
CD	*7.225E-003*	*1*	*7.225E-003*	*4.52*	*0.0475*	
A2	*7.511E-003*	*1*	*7.511E-003*	*4.70*	*0.0438*	
C2	*0.013*	*1*	*0.013*	*8.04*	*0.0110*	
Residual	*0.029*	*18*	*1.597E-003*			
Lack of Fit	*0.020*	*13*	*1.513E-003*	*0.83*	*0.6382*	*not significant*
Pure Error	*9.083E-003*	*5*	*1.817E-003*			
Cor Total	*0.31*	*29*				

Std. Dev.	0.040	R-Squared	0.9060
Mean	3.66	Adj R-Squared	0.8486
C.V. %	1.09	Pred R-Squared	0.7581
PRESS	0.074	Adeq Precision	19.913

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	3.70	1	0.012	3.67	3.72	
A-Wing Area	-8.333E-004	1	8.158E-003	-0.018	0.016	1.00
B-Length/Width	0.051	1	8.158E-003	0.034	0.068	1.00
C-Base Width	2.500E-003	1	8.158E-003	-0.015	0.020	1.00
D-Base Length	-0.061	1	8.158E-003	-0.078	-0.044	1.00
AB	-0.029	1	9.991E-003	-0.050	-7.759E-003	1.00
AC	-0.037	1	9.991E-003	-0.058	-0.017	1.00
AD	0.044	1	9.991E-003	0.023	0.065	1.00
BC	0.046	1	9.991E-003	0.025	0.067	1.00
CD	-0.021	1	9.991E-003	-0.042	-2.590E-004	1.00
A2	-0.016	1	7.493E-003	-0.032	-5.068E-004	1.01
C2	-0.021	1	7.493E-003	-0.037	-5.507E-003	1.01

Final Equation in Terms of Actual Factors:

Avg Flt Time	=
+3.70	
-8.333E-004	* A
+0.051	* B
+2.500E-003	* C
-0.061	* D
-0.029	* A * B
-0.037	* A * C
+0.044	* A * D

+0.046	* B * C
-0.021	* C * D
-0.016	* A2
-0.021	* C2

Final Equation in Terms of Actual Factors:

Avg Flt Time	=
-8.13881	
+1.59365	* Wing Area
+1.56132	* Length/Width Ratio
+2.54425	* Base Width
-1.71750	* Base Length
-0.18082	* Wing Area * Length/Width Ratio
-0.25000	* Wing Area * Base Width
+0.14583	* Wing Area * Base Length
+0.69811	* Length/Width Ratio * Base Width
-0.17000	* Base Width * Base Length
-0.045139	* Wing Area2
-0.34000	* Base Width2

(b) Fit a second-order model to the standard deviation of flight time response.

The standard deviation response is driven by the *CD* interaction and A^2. *A*, C and *D* are in the model to maintain hierarchy.

Design Expert Output

Response 2 Stdev Flt Time

ANOVA for Response Surface Reduced Quadratic Model

Analysis of variance table [Partial sum of squares - Type III]

Source	**Sum of Squares**	**df**	**Mean Square**	**F Value**	**p-value Prob > F**	
Model	5.562E-003	5	1.112E-003	2.57	0.0533	not significant
A-Wing Area	*1.504E-005*	*1*	*1.504E-005*	*0.035*	*0.8536*	
C-Base Width	*2.042E-006*	*1*	*2.042E-006*	*4.72E-003*	*0.9458*	
D-Base Length	*6.934E-004*	*1*	*6.934E-004*	*1.60*	*0.2176*	
CD	*1.871E-003*	*1*	*1.871E-003*	*4.32*	*0.0484*	
A2	*2.981E-003*	*1*	*2.981E-003*	*6.89*	*0.0148*	
Residual	0.010	24	4.325E-004			
Lack of Fit	*9.091E-003*	*19*	*4.785E-004*	*1.86*	*0.2556*	*not significant*
Pure Error	*1.289E-003*	*5*	*2.579E-004*			
Cor Total	0.016	29				

Std. Dev.	0.021	R-Squared	0.3489
Mean	0.069	Adj R-Squared	0.2132
C.V. %	30.14	Pred R-Squared	-0.1717
PRESS	0.019	Adeq Precision	5.702

Factor	**Coefficient Estimate**	**df**	**Standard Error**	**95% CI Low**	**95% CI High**	**VIF**
Intercept	0.061	1	4.90E-003	0.051	0.071	
A-Wing Area	-7.917E-004	1	4.24E-003	-9.55E-003	7.970E-003	1.00
C-Base Width	2.917E-004	1	4.24E-003	-8.47E-003	9.053E-003	1.00
D-Base Length	5.375E-003	1	4.24E-003	-3.38E-003	0.014	1.00
CD	-0.011	1	5.19E-003	-0.022	-8.191E-005	1.00
A2	0.010	1	3.87E-003	2.175E-003	0.018	1.00

Final Equation in Terms of Coded Factors:

Stdev Flt Time	=
+0.061	
-7.917E-004	* A
+2.917E-004	* C
+5.375E-003	* D

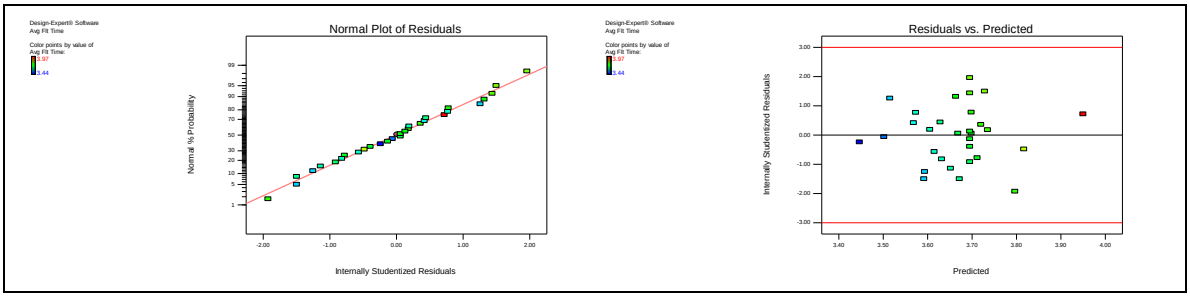
-0.011 * C * D
+0.010 * A2

Final Equation in Terms of Actual Factors:

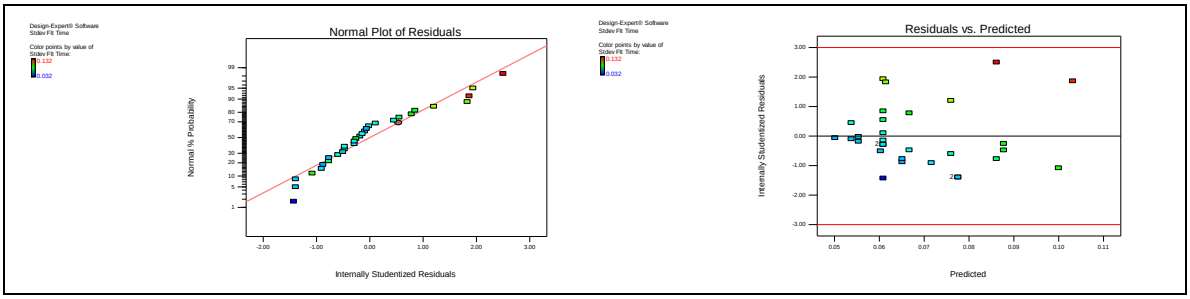
Stdev Flt Time =
+4.18328
-0.70217 * Wing Area
+0.17417 * Base Width
+0.11887 * Base Length
-0.086500 * Base Width * Base Length
+0.028260 * Wing Area2

(c) Analyze the residuals for both models from parts (a) and (b). Are transformations on the response(s) necessary? If so, fit the appropriate models.

For the average flight time response the residuals look fine. The residual by factor are also okay.



For the standard deviation of flight time response the plot of residuals vs. predicted shows an inequality of variance. Recommend doing a Log transformation.



Design Expert Output

Response 2 Stdev Flt Time
Transform: Base 10 Log Constant: 0

ANOVA for Response Surface Reduced Quadratic Model
Analysis of variance table [Partial sum of squares - Type III]

Source	**Sum of Squares**	**df**	**Mean Square**	**F Value**	**p-value Prob > F**	
Model	0.18	5	0.035	2.34	0.0725	not sig
A-Wing Area	*1.089E-004*	*1*	*1.089E-004*	*7.212E-003*	*0.9330*	
C-Base Width	*3.136E-004*	*1*	*3.136E-004*	*0.021*	*0.8866*	
D-Base Length	*0.024*	*1*	*0.024*	*1.61*	*0.2167*	
CD	*0.062*	*1*	*0.062*	*4.11*	*0.0539*	
A2	*0.090*	*1*	*0.090*	*5.96*	*0.0224*	
Residual	0.36	24	0.015			
Lack of Fit	*0.27*	*19*	*0.014*	*0.77*	*0.6920*	*not sig*
Pure Error	*0.092*	*5*	*0.018*			
Cor Total	0.54	29				

Std. Dev.	0.12	R-Squared	0.3278
Mean	-1.18	Adj R-Squared	0.1878
C.V. %	10.39	Pred R-Squared	-0.0669
PRESS	0.58	Adeq Precision	5.304

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	-1.23	1	0.029	-1.29	-1.17	
A-Wing Area	-2.130E-003	1	0.025	-0.054	0.050	1.00
C-Base Width	3.615E-003	1	0.025	-0.048	0.055	1.00
D-Base Length	0.032	1	0.025	-0.020	0.084	1.00
CD	-0.062	1	0.031	-0.13	1.13E-003	1.00
A2	0.056	1	0.023	8.64E-003	0.10	1.00

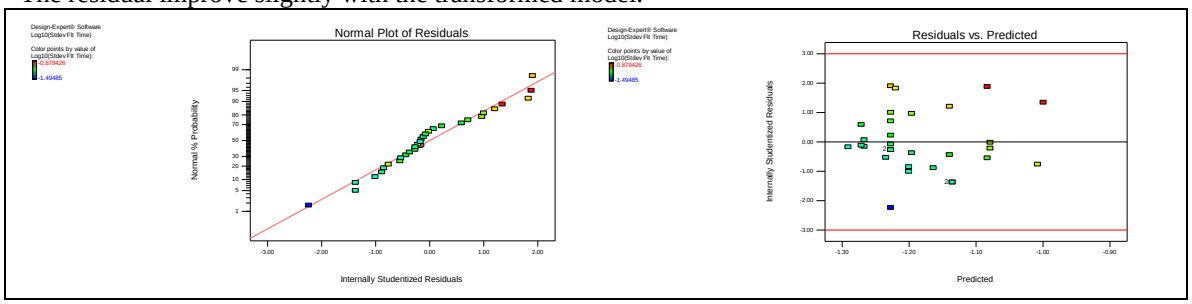
Final Equation in Terms of Coded Factors:

Log10(Stdev Flt Time) =
-1.23
-2.130E-003 * A
+3.615E-003 * C
+0.032 * D
-0.062 * C * D
+0.056 * A2

Final Equation in Terms of Actual Factors:

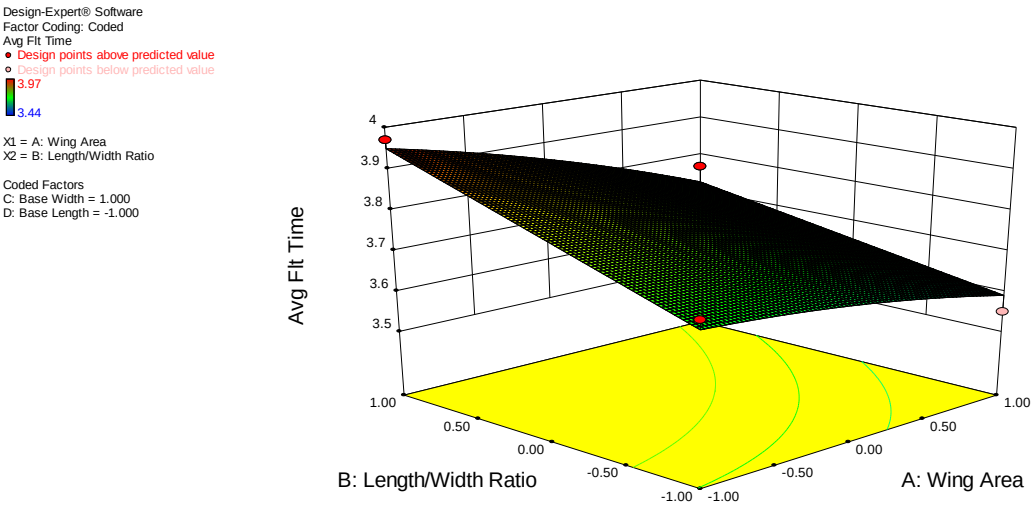
Log10(Stdev Flt Time) =
+21.30149
-3.85446 * Wing Area
+1.01078 * Base Width
+0.68634 * Base Length
-0.49816 * Base Width * Base Length
+0.15528 * Wing Area2

The residual improve slightly with the transformed model.



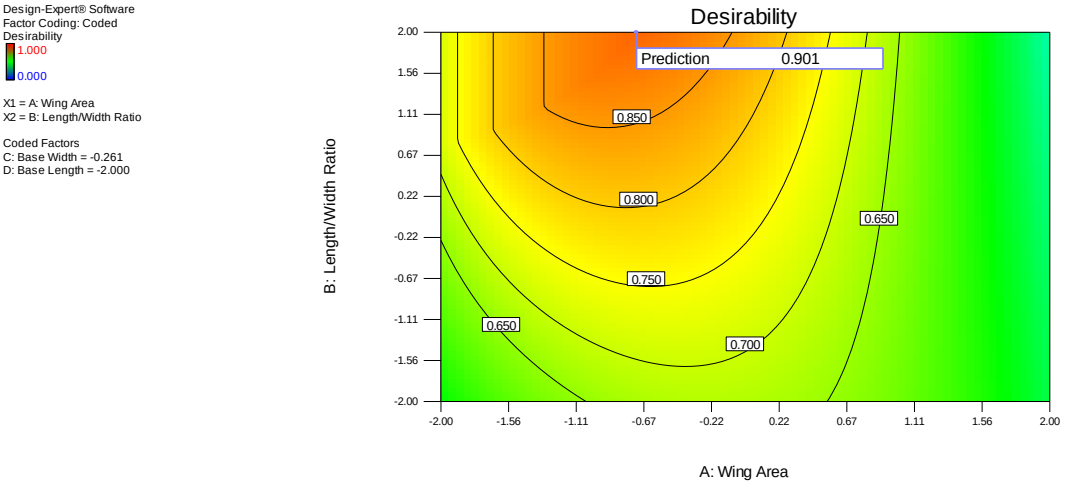
(d) What design would you recommend to maximize the flight time?

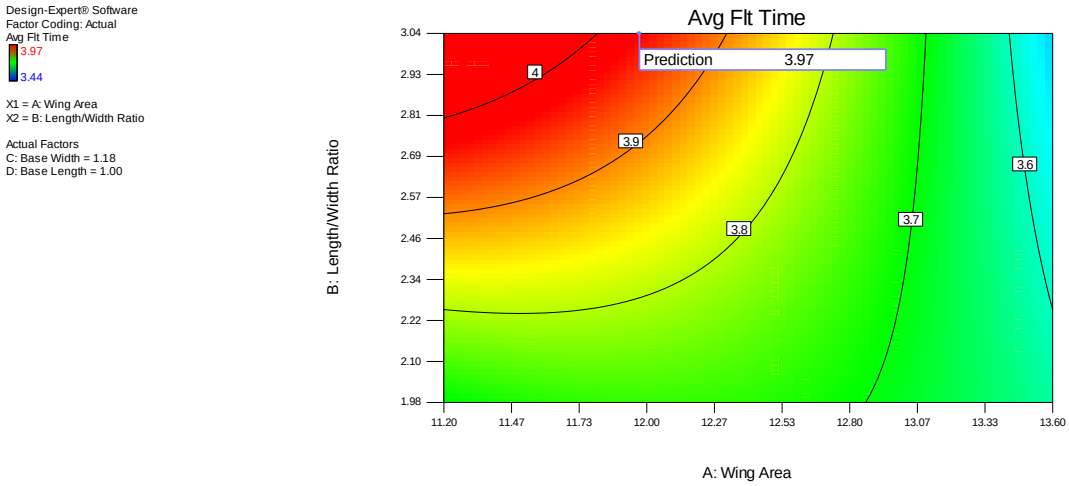
Run *A* at the low level, *B* at the high level, *C* at the high level and *D* at the low level. This give a flight time of almost 4 seconds.



(e) What design would you recommend to maximize the flight time while simultaneously minimizing the standard deviation of flight time?

Running *A* at -0.715, *B* at -2, *C* at -0.261 and *D* at -2, in coded terms. Applying these values to the regression equations estimates the hang time of 3.97 seconds and the standard deviation at 0.051.

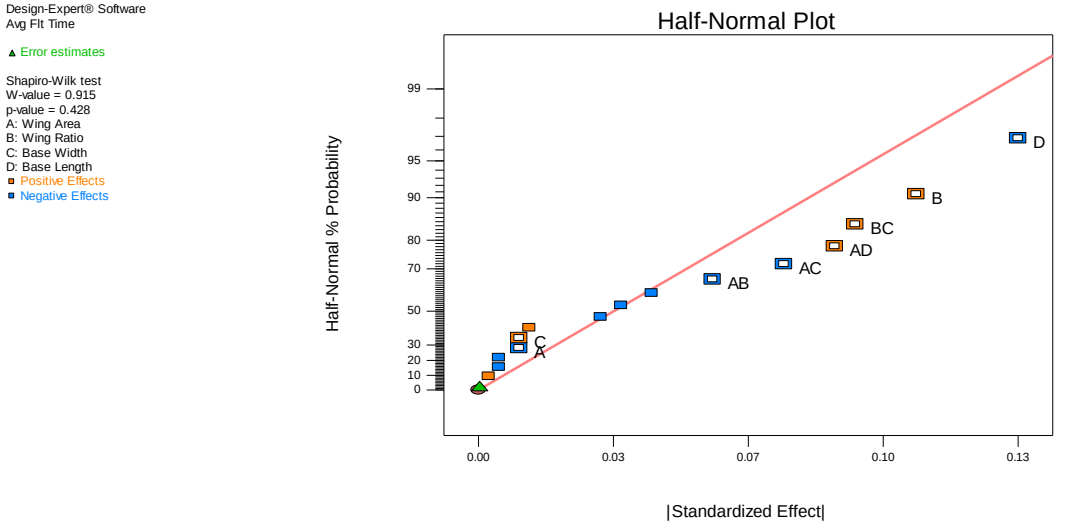




11.29. The Paper Helicopter Experiment Revisited. Reconsider the paper helicopter experiment in Problem 11.28. This experiment was actually run in two blocks. Block 1 consisted of the first 16 runs in Table P11.11 (standard order runs 1-16) and two center points (standard order runs 25 and 26).

(a) Fit the main-effects plus two-factor interaction models to the block 1 data, using both responses.

For the average flight time: The main effects of *B* and *D* are significant as well as the two-factor interactions of *AB*, *AC*, *AD*, *BC*. *A* and *C* are added to maintain hierarchy.



Design Expert Output

ANOVA for selected factorial model
Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	**0.22**	**8**	**0.027**	**12.99**	**0.0008**	**significant**
A-Wing Area	*4.00E-004*	*1*	*4.00E-004*	*0.19*	*0.6738*	
B-Wing Ratio	*0.046*	*1*	*0.046*	*22.04*	*0.0016*	
C-Base Width	*4.00E-004*	*1*	*4.00E-004*	*0.19*	*0.6738*	
D-Base Length	*0.070*	*1*	*0.070*	*33.49*	*0.0004*	

AB	*0.013*	*1*	*0.013*	*6.31*	*0.0363*	
AC	*0.022*	*1*	*0.022*	*10.73*	*0.0113*	
AD	*0.031*	*1*	*0.031*	*14.61*	*0.0051*	
BC	*0.034*	*1*	*0.034*	*16.32*	*0.0037*	
Curvature	0.018	1	0.018	8.48	0.0195	significant
Residual	0.017	8	2.09E-003			
Lack of Fit	*0.017*	*7*	*2.36E-003*	*11.84*	*0.2202*	*not significant*
Pure Error	*2.00E-004*	*1*	*2.00E-004*			
Cor Total	0.25	17				

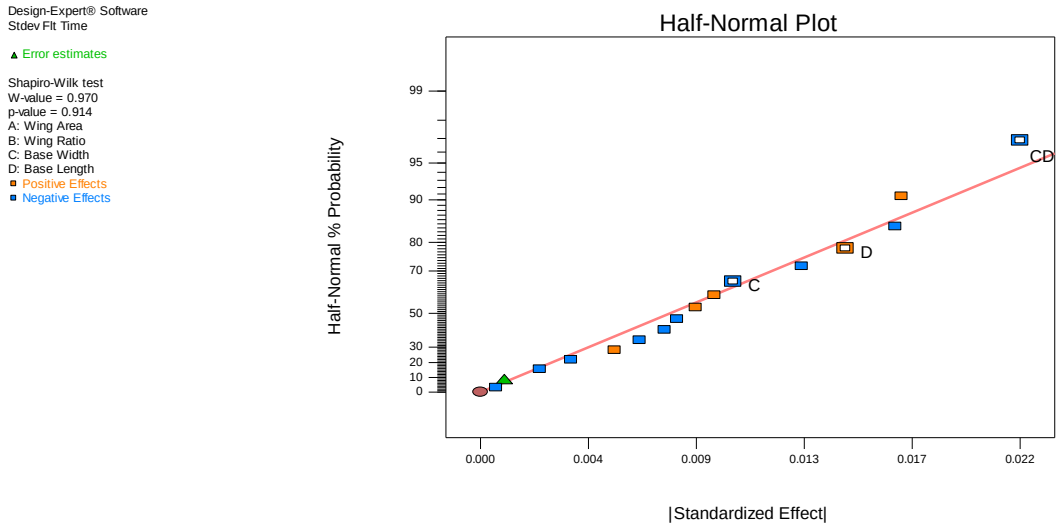
Std. Dev.	0.062	R-Squared	0.8631
Mean	3.67	Adj R-Squared	0.7414
C.V. %	1.69	Pred R-Squared	0.5568
PRESS	0.11	Adeq Precision	10.899

Factor	**Coefficient Estimate**	**df**	**Standard Error**	**95% CI Low**	**95% CI High**	**VIF**
Intercept	3.67	1	0.015	3.64	3.70	
A-Wing Area	-5.00E-003	1	0.015	-0.040	0.030	1.00
B-Wing Ratio	0.054	1	0.015	0.019	0.089	1.00
C-Base Width	5.00E-003	1	0.015	-0.030	0.040	1.00
D-Base Length	-0.066	1	0.015	-0.10	-0.031	1.00
AB	-0.029	1	0.015	-0.064	6.29E-003	1.00
AC	-0.037	1	0.015	-0.073	-2.46E-003	1.00
AD	0.044	1	0.015	8.71E-003	0.079	1.00
BC	0.046	1	0.015	0.011	0.081	1.00

Final Equation in Terms of Coded Factors:

Avg Flt Time =
+3.67
-5.00E-003 * A
+0.054 * B
+5.00E-003 * C
-0.066 * D
-0.029 * A * B
-0.037 * A * C
+0.044 * A * D
+0.046 * B * C

For the Standard Deviation Flight Time response: Only the *CD* interaction is significant. *C* and *D* are added to maintain hierarchy.



Design Expert Output

ANOVA for selected factorial model
Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	3.136E-003	3	1.045E-003	2.93	0.0735	not significant
C-Base Width	*4.101E-004*	*1*	*4.101E-004*	*1.15*	*0.3034*	
D-Base Length	*8.556E-004*	*1*	*8.556E-004*	*2.40*	*0.1456*	
CD	*1.871E-003*	*1*	*1.871E-003*	*5.24*	*0.0395*	
Curvature	1.095E-003	1	1.095E-003	3.07	0.1035	not significant
Residual	4.642E-003	13	3.571E-004			
Lack of Fit	*4.377E-003*	*12*	*3.648E-004*	*1.38*	*0.5889*	*not significant*
Pure Error	*2.645E-004*	*1*	*2.645E-004*			
Cor Total	8.872E-003	17				

Std. Dev.	0.020	R-Squared	0.3535
Mean	0.066	Adj R-Squared	0.2149
C.V. %	30.88	Pred R-Squared	-0.0413
PRESS	9.239E-003	Adeq Precision	3.799

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	0.066	1	4.771E-003	0.055	0.076	
C-Base Width	-5.063E-003	1	5.060E-003	-0.016	5.791E-003	1.00
D-Base Length	7.312E-003	1	5.060E-003	-3.54E-003	0.018	1.00
CD	-0.011	1	5.060E-003	-0.022	4.111E-005	1.00

Final Equation in Terms of Coded Factors:

Stdev Flt Time =
+0.066
-5.063E-003 * C
+7.312E-003 * D
-0.011 * C * D

(b) For the models in part (a) use the two center points to test for lack of fit. Is there an indication that second-order terms are needed?

The ANOVA for Average Flight Time shows curvature to be significant. The ANOVA for Standard Deviation Flight Time does not show curvature.

(c) Now use the data from block 2 (standard order runs 17-24 and the remaining center points, standard order runs 27-30) to augment block 1 and fit second-order models to both responses. Check the adequacy of the fit for both models. Does blocking seem to have been important in this experiment?

Blocking has little effect on the models.

For the average flight time of the combined design: The main effects of *B* and *D* are significant as well as the two-factor interactions of *AB*, *AC*, *AD*, *BC* and *CD*. The quadratic terms of A^2, B^2 and C^2 are significant. *A* and *C* are added to maintain hierarchy.

Design Expert Output

Response 1 Avg Flt Time
ANOVA for Response Surface Reduced Quadratic Model
Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Block	1.68E-003	1	1.681E-003			
Model	0.29	12	0.024	22.14	< 0.0001	significant
A-Wing Area	*1.67E-005*	*1*	*1.667E-005*	*0.015*	*0.9027*	
B-Wing Ratio	*0.062*	*1*	*0.062*	*57.40*	*< 0.0001*	
C-Base Width	*1.50E-004*	*1*	*1.500E-004*	*0.14*	*0.7143*	
D-Base Length	*0.089*	*1*	*0.089*	*82.21*	*< 0.0001*	

AB	*0.013*	*1*	*0.013*	*12.24*	*0.0030*	
AC	*0.022*	*1*	*0.022*	*20.83*	*0.0003*	
AD	*0.031*	*1*	*0.031*	*28.35*	*< 0.0001*	
BC	*0.034*	*1*	*0.034*	*31.68*	*< 0.0001*	
CD	*7.22E-003*	*1*	*7.225E-003*	*6.69*	*0.0199*	
A2	*0.011*	*1*	*0.011*	*10.26*	*0.0055*	
B2	*7.33E-003*	*1*	*7.336E-003*	*6.79*	*0.0191*	
C2	*0.017*	*1*	*0.017*	*16.00*	*0.0010*	
Residual	0.017	16	1.080E-003			
Lack of Fit	*0.016*	*12*	*1.351E-003*	*5.03*	*0.0659*	*not significant*
Pure Error	*1.07E-003*	*4*	*2.688E-004*			
Cor Total	0.31	29				

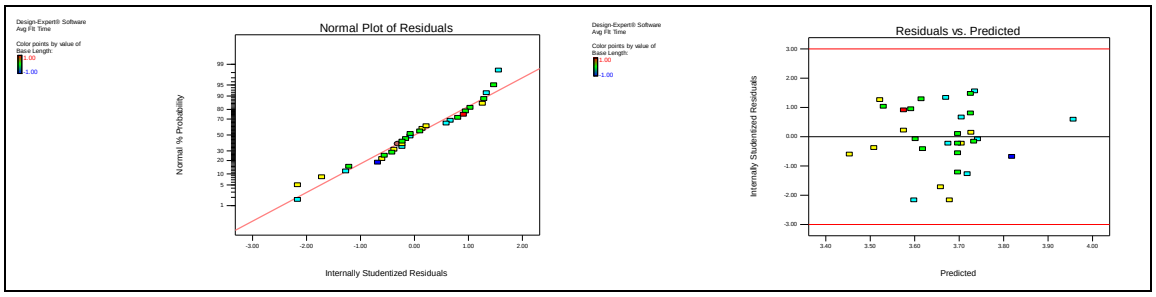
Std. Dev.	0.033	R-Squared	0.9432
Mean	3.66	Adj R-Squared	0.9006
C.V. %	0.90	Pred R-Squared	0.7728
PRESS	0.069	Adeq Precision	22.417

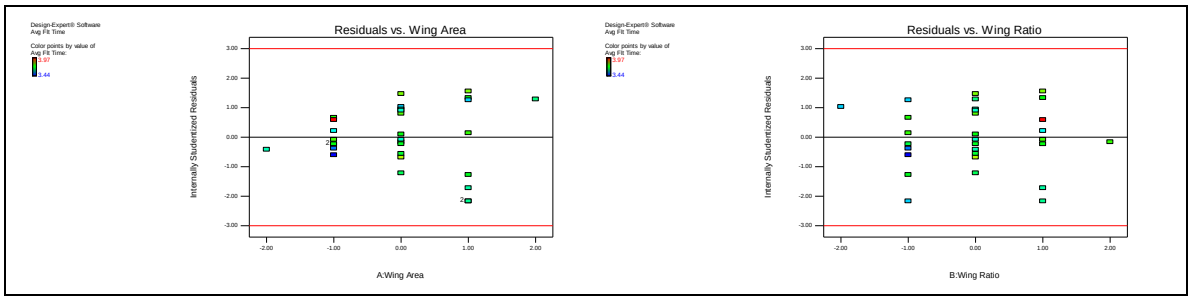
Factor	**Coefficient Estimate**	**df**	**Standard Error**		**95% CI Low**	**95% CI High**	**VIF**
Intercept	3.71	1	0.011	3	.69	3.74	
Block 1	0.014	1					
Block 2	-0.014						
A-Wing Area	-8.33E-004	1	6.709E-003		-0.015	0.013	1.00
B-Wing Ratio	0.051	1	6.709E-003		0.037	0.065	1.00
C-Base Width	2.50E-003	1	6.709E-003		-0.012	0.017	1.00
D-Base Length	-0.061	1	6.709E-003		-0.075	-0.047	1.00
AB	-0.029	1	8.217E-003		-0.046	-0.011	1.00
AC	-0.037	1	8.217E-003		-0.055	-0.020	1.00
AD	0.044	1	8.217E-003		0.026	0.061	1.00
BC	0.046	1	8.217E-003		0.029	0.064	1.00
CD	-0.021	1	8.217E-003		-0.039	-3.830E-003	1.00
A2	-0.020	1	6.276E-003		-0.033	-6.800E-003	1.05
B2	-0.016	1	6.276E-003		-0.030	-3.050E-003	1.05
C2	-0.025	1	6.276E-003		-0.038	-0.012	1.05

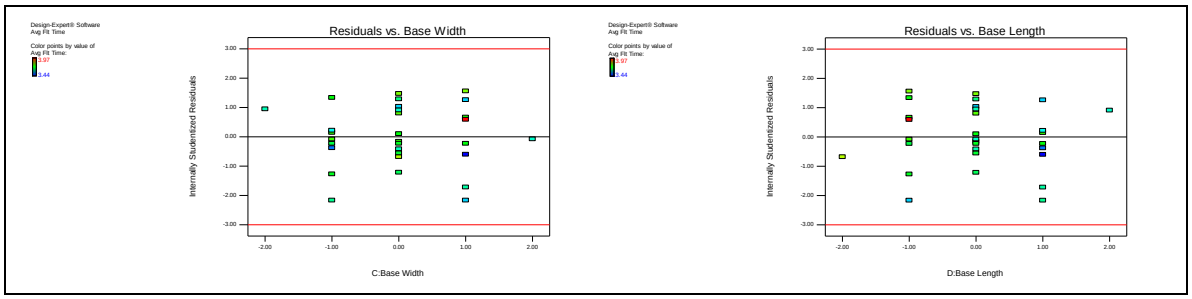
Final Equation in Terms of Coded Factors:

Avg Flt Time	=
+3.71	
-8.333E-004	* A
+0.051	* B
+2.500E-003	* C
-0.061	* D
-0.029	* A * B
-0.037	* A * C
+0.044	* A * D
+0.046	* B * C
-0.021	* C * D
-0.020	* A2
-0.016	* B2
-0.025	* C2

The residual analysis for the Average Flight Time response shows a little inequality of variance in Factor A.







For the Standard Deviation Flight Time response in the combined model: Only the *CD* interaction and the A^2 term are significant. *A*, *C* and *D* are added to maintain hierarchy.

Design Expert Output

Response 2 Stdev Flt Time

ANOVA for Response Surface Reduced Quadratic Model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Block	5.339E-004	1	5.339E-004			
Model	5.890E-003	5	1.178E-003	2.85	0.0383	significant
A-Wing Area	*1.504E-005*	*1*	*1.504E-005*	*0.036*	*0.8505*	
C-Base Width	*2.042E-006*	*1*	*2.042E-006*	*4.93E-003*	*0.9446*	
D-Base Length	*6.934E-004*	*1*	*6.934E-004*	*1.68*	*0.2084*	
CD	*1.871E-003*	*1*	*1.871E-003*	*4.52*	*0.0445*	
A2	*3.309E-003*	*1*	*3.309E-003*	*7.99*	*0.0095*	
Residual	9.518E-003	23	4.138E-004			
Lack of Fit	*8.889E-003*	*19*	*4.679E-004*	*2.97*	*0.1499*	*not significant*
Pure Error	*6.292E-004*	*4*	*1.573E-004*			
Cor Total	0.016	29				

Std. Dev.	0.020	R-Squared	0.3822
Mean	0.069	Adj R-Squared	0.2479
C.V. %	29.48	Pred R-Squared	-0.3520
PRESS	0.021	Adeq Precision	6.331

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	0.061	1	4.814E-003	0.052	0.071	

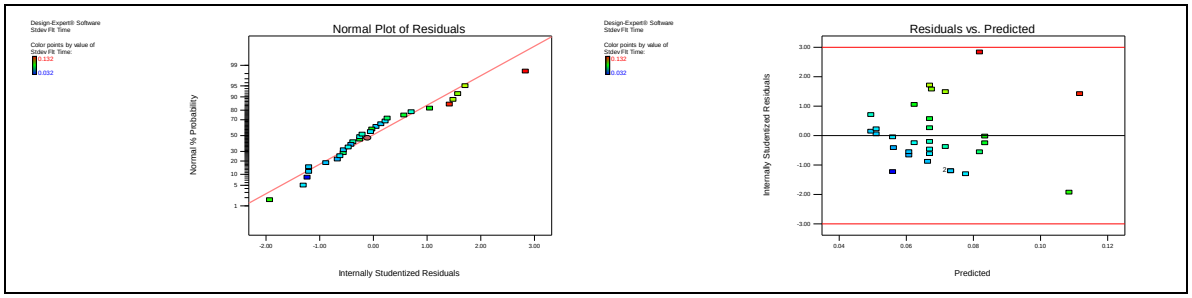
Block 1	-5.50E-003	1				
Block 2	5.50E-003					
A-Wing Area	-7.92E-004	1	4.153E-003	-9.382E-003	7.799E-003	1.00
C-Base Width	2.92E-004	1	4.153E-003	-8.299E-003	8.882E-003	1.00
D-Base Length	5.37E-003	1	4.153E-003	-3.215E-003	0.014	1.00
CD	-0.011	1	5.086E-003	-0.021	-2.917E-004	1.00
A2	0.011	1	3.814E-003	2.895E-003	0.019	1.01

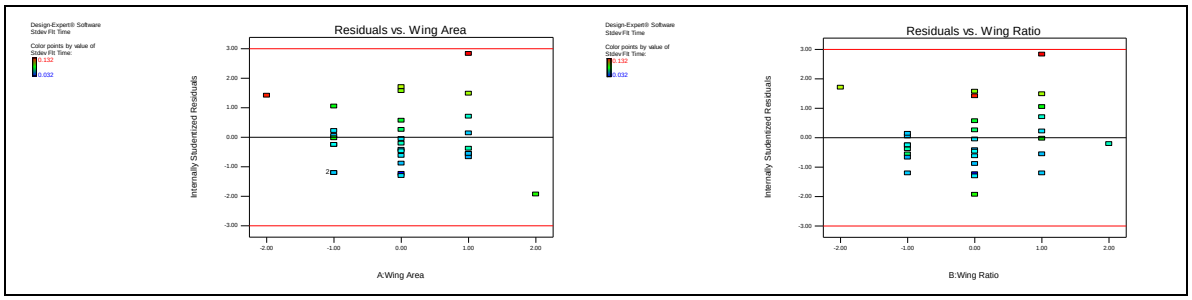
Final Equation in Terms of Coded Factors:

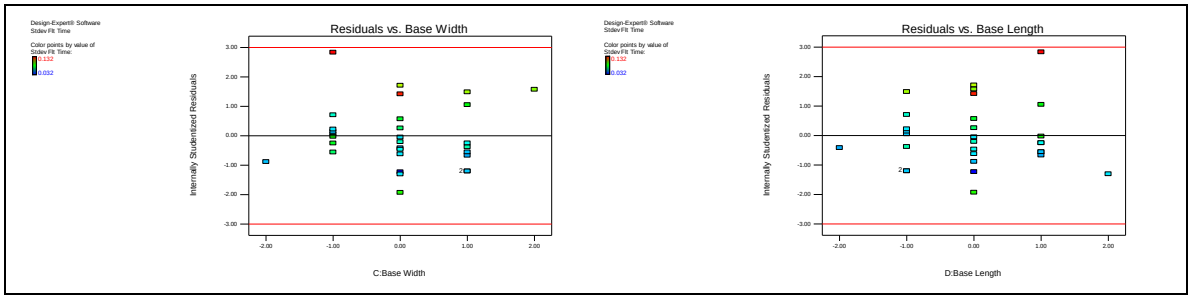
Stdev Flt Time =

+0.061
-7.917E-004 * A
+2.917E-004 * C
+5.375E-003 * D
-0.011 * C * D
+0.011 * A2

The residual analysis of the Standard Deviation Flight Time response shows in equality of variance in the Residuals vs. Predicted graph and in Factor *B*. Might want to try a transformation on the Standard Deviation Response.

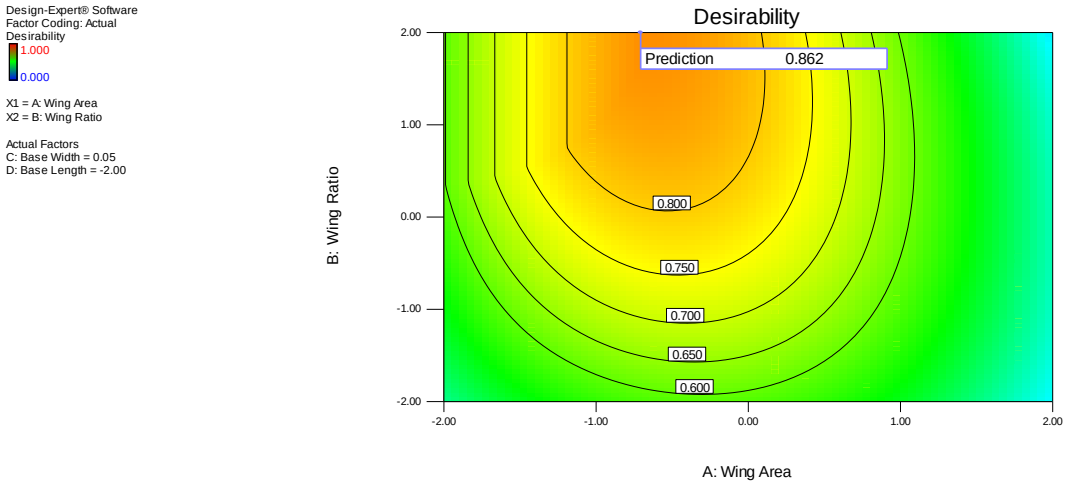


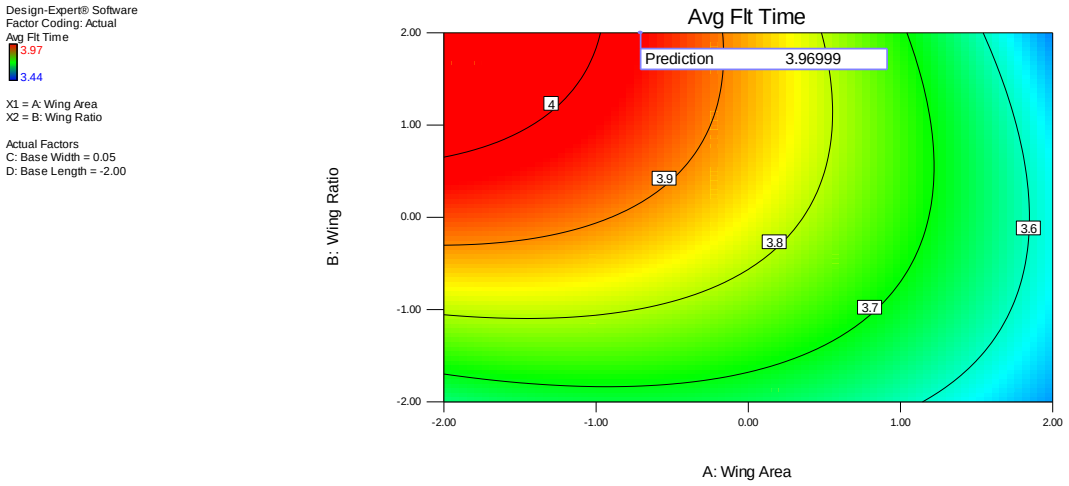




(d) What design would you recommend to maximize the flight time while simultaneously minimizing the standard deviation of flight time?

A = -0.71, B = 2, C = 0.05, D = -2, in coded variables gives a predicted Flight Time of 3.967 and a predicted Standard Deviation of Flight Time of 0.058.





11.30. An article in the *Journal of Chromatography A* ("Optimization of the Capillary Electrophoresis Separation of Ranitidine and Related Compounds," Vol. 766, pp. 245-254) describes an experiment to optimize the production of ranitidine, a compound that is the primary active ingredient of Zantac, a pharmaceutical product used to treat ulcers, gastroesophageal reflux disease (a condition in which backward flow of acid from the stomach causes heartburn and injury of the esophagus), and other conditions where the stomach produces too much acid, such as Zollinger-Ellison syndrome. The authors used three factors (x_1 = pH of the buffer solution, x_2 = the electrophoresis voltage, and the concentration of one component of the buffer solution) in a central composite design. The response is chromatographic exponential function (CEF), which should be minimized. Table P11.12 shows the design.

P11.12 – The Ranitidine Separation Experiment

Std Ord	*X*1	*X*2	*X*3	CEF
1	-1	-1	-1	17.3
2	1	-1	-1	45.5
3	-1	1	-1	10.3
4	1	1	-1	11757.1
5	-1	-1	1	16.942

6	1	-1	1	25.4
7	-1	1	1	31697.2
8	1	1	1	12039.2
9	-1.68	0	0	7.5
10	1.68	0	0	6.3
11	0	-1.68	0	11.1
12	0	1.68	0	6.6664
13	0	0	-1.68	16548.7
14	0	0	1.68	26351.8
15	0	0	0	9.9
16	0	0	0	9.6
17	0	0	0	8.9
18	0	0	0	8.8
19	0	0	0	8.013
20	0	0	0	8.059

(a) Fit the second-order model to the CEF response. Analyze the residuals from this model. Does it seem that all model terms are necessary?

There are many terms with low *F*-Values. The Adjusted R-Square is low relative to the *R*-Square indicating too many terms in the model.

Design Expert Output

Response 1 CEF

ANOVA for Response Surface Quadratic Model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	1.480E+009	9	1.644E+008	5.66	0.0061	significant
A-pH	*4.543E+006*	*1*	*4.543E+006*	*0.16*	*0.7008*	
B-voltage	*2.247E+008*	*1*	*2.247E+008*	*7.73*	*0.0194*	
C-concen	*1.718E+008*	*1*	*1.718E+008*	*5.91*	*0.0353*	
AB	*7.896E+006*	*1*	*7.896E+006*	*0.27*	*0.6134*	
AC	*1.234E+008*	*1*	*1.234E+008*	*4.25*	*0.0662*	
BC	*1.279E+008*	*1*	*1.279E+008*	*4.40*	*0.0622*	
A2	*2.637E+005*	*1*	*2.637E+005*	*9.080E-003*	*0.9260*	
B2	*2.610E+005*	*1*	*2.610E+005*	*8.986E-003*	*0.9264*	
C2	*7.990E+008*	*1*	*7.990E+008*	*27.51*	*0.0004*	
Residual	2.905E+008	10	2.905E+007			
Lack of Fit	*2.905E+008*	*5*	*5.809E+007*	*9.710E+007*	*< 0.0001*	*significant*
Pure Error	*2.99*	*5*	*0.60*			
Cor Total	1.770E+009	19				

Std. Dev.	5389.41	R-Squared	0.8359
Mean	4929.71	Adj R-Squared	0.6882
C.V. %	109.32	Pred R-Squared	-0.3559
PRESS	2.40E+009	Adeq Precision	8.981

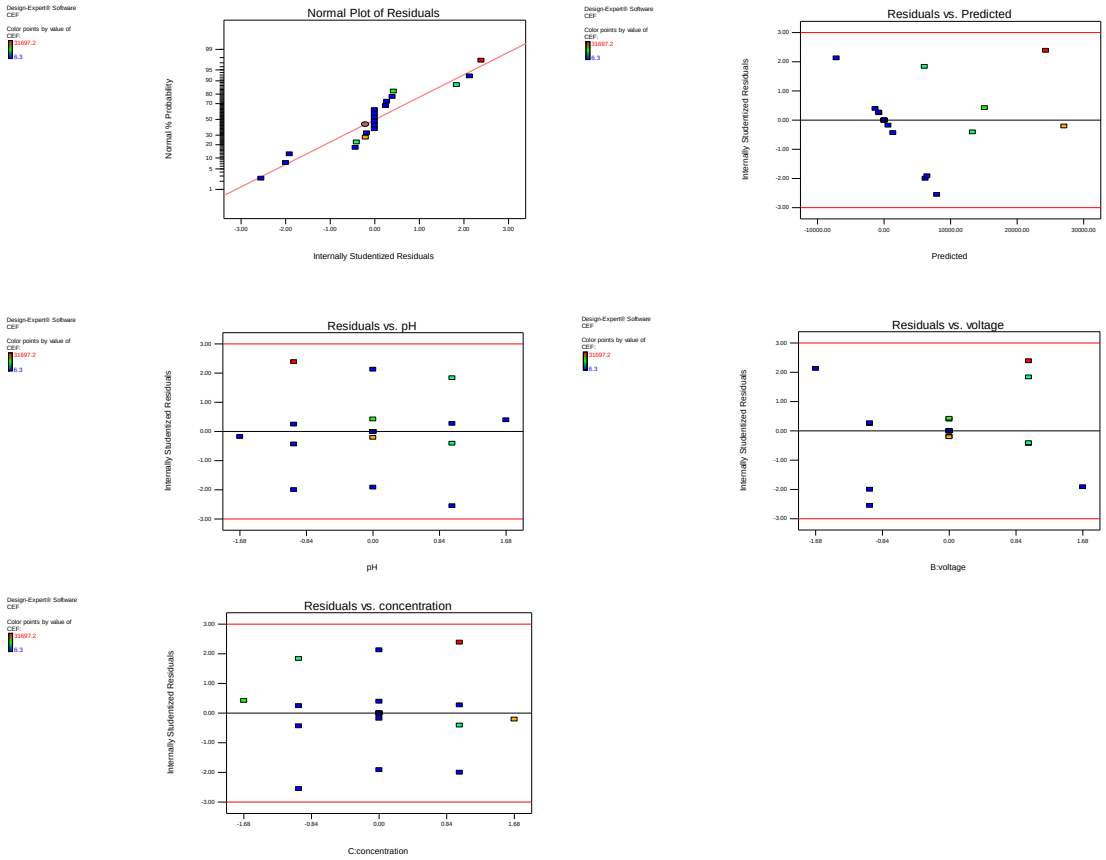
Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	29.47	1	2198.06	-4868.12	4927.06	
A-pH	-576.75	1	1458.36	-3826.18	2672.69	1.00
B-voltage	4055.93	1	1458.36	806.49	7305.36	1.00
C-concentration	3546.59	1	1458.36	297.16	6796.03	1.00
AB	-993.48	1	1905.44	-5239.08	3252.11	1.00
AC	-3928.07	1	1905.44	-8173.66	317.53	1.00
BC	3998.68	1	1905.44	-246.91	8244.28	1.00
A2	-135.28	1	1419.68	-3298.51	3027.96	1.02
B2	-134.58	1	1419.68	-3297.81	3028.66	1.02
C2	7446.09	1	1419.68	4282.85	10609.33	1.02

Final Equation in Terms of Coded Factors:

CEF =

```
+29.47
-576.75     * A
+4055.93    * B
+3546.59    * C
-993.48     * A * B
-3928.07    * A * C
+3998.68    * B * C
-135.28     * A2
-134.58     * B2
+7446.09    * C2
```

The Normal Plot of residuals is somewhat S-shaped.



(b) Reduce the model from part (a) as necessary. Did the model reduction improve the fit?

The Adjusted *R*-Square improved. The only terms left in the model have significant F-Values.

Design Expert Output

Response 1 CEF

ANOVA for Response Surface Reduced Quadratic Model

Analysis of variance table [Partial sum of squares - Type III]

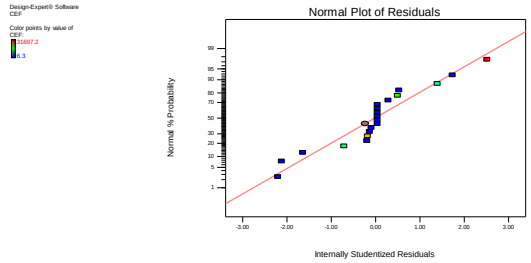
Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	1.471E+009	6	2.452E+008	10.67	0.0002	significant
A-pH	*4.543E+006*	*1*	*4.543E+006*	*0.20*	*0.6640*	
B-voltage	*2.247E+008*	*1*	*2.247E+008*	*9.77*	*0.0080*	
C-concen	*1.718E+008*	*1*	*1.718E+008*	*7.47*	*0.0171*	
AC	*1.234E+008*	*1*	*1.234E+008*	*5.37*	*0.0374*	
BC	*1.279E+008*	*1*	*1.279E+008*	*5.56*	*0.0346*	
C2	*8.190E+008*	*1*	*8.190E+008*	*35.63*	*< 0.0001*	

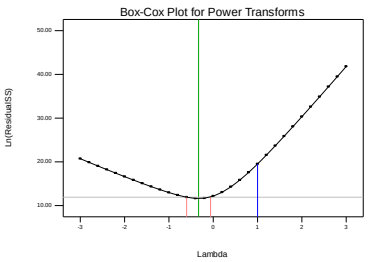
Residual	2.988E+008	13	2.299E+007			
Lack of Fit	*2.988E+008*	*8*	*3.735E+007*	*6.244E+007*	*< 0.0001*	*significant*
Pure Error	*2.99*	*5*	*0.60*			
Cor Total	1.770E+009	19				

Std. Dev.	4794.47	R-Squared	0.8312
Mean	4929.71	Adj R-Squared	0.7533
C.V. %	97.26	Pred R-Squared	0.3793
PRESS	1.099E+009	Adeq Precision	11.957

(c) Does transformation of the CEF response seem like a useful idea? What aspect of either the data or the residuals suggests that transformation would be helpful?

The Normal Plot of Residuals still shows an S-shaped pattern in the reduced model. Suggest an inverse square root transformation.





(d) Fit a second-order model to the transformed CEF response. Analyze the residuals from this model. Does it seem that all model terms are necessary? What would you choose as the final model?

Like the first model, there are too many terms. Many with low *F*-Values that could be deleted from the model.

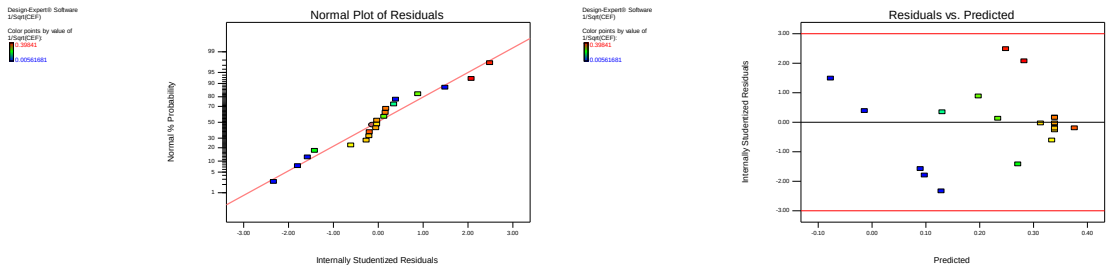
Design Expert Output

Response 1 CEF
Transform: Inverse Sqrt Constant: 0
ANOVA for Response Surface Quadratic Model
Analysis of variance table [Partial sum of squares - Type III]

Source	**Sum of Squares**	**df**	**Mean Square**	**F Value**	**p-value Prob > F**	
Model	0.33	9	0.036	4.59	0.0130	significant
A-pH	*0.011*	*1*	*0.011*	*1.33*	*0.2757*	
B-voltage	*8.864E-003*	*1*	*8.864E-003*	*1.12*	*0.3155*	
C-concen	*4.802E-003*	*1*	*4.802E-003*	*0.60*	*0.4547*	
AB	*3.287E-003*	*1*	*3.287E-003*	*0.41*	*0.5344*	
AC	*0.016*	*1*	*0.016*	*1.97*	*0.1910*	
BC	*0.016*	*1*	*0.016*	*2.03*	*0.1850*	
A2	*1.819E-004*	*1*	*1.819E-004*	*0.023*	*0.8827*	
B2	*4.168E-003*	*1*	*4.168E-003*	*0.52*	*0.4853*	
C2	*0.27*	*1*	*0.27*	*33.61*	*0.0002*	
Residual	0.079	10	7.939E-003			
Lack of Fit	*0.078*	*5*	*0.016*	*73.24*	*0.0001*	*significant*
Pure Error	*1.069E-003*	*5*	*2.139E-004*			
Cor Total	0.41	19				

Std. Dev.	0.089	R-Squared	0.8050
Mean	0.23	Adj R-Squared	0.6295
C.V. %	38.33	Pred R-Squared	-0.4939
PRESS	0.61	Adeq Precision	7.192

The model residuals have improved.



The reduced model includes only factor C and C^2.

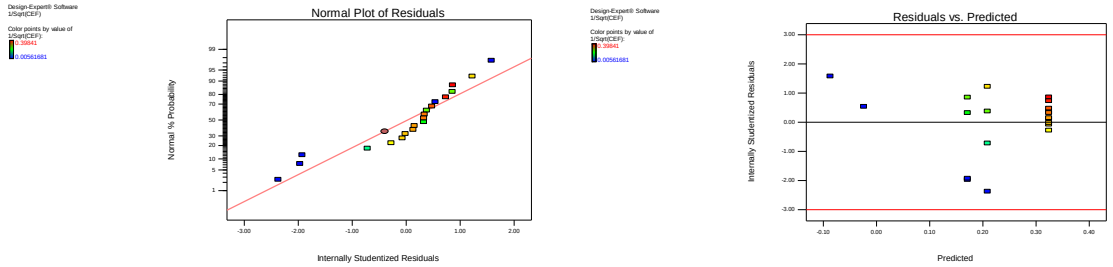
Design Expert Output

Response 1 CEF
Transform: Inverse Sqrt Constant: 0

ANOVA for Response Surface Reduced Quadratic Model
Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	0.27	2	0.13	16.57	0.0001	significant
C-conce	*4.802E-003*	*1*	*4.802E-003*	*0.59*	*0.4524*	
C2	*0.26*	*1*	*0.26*	*32.56*	*< 0.0001*	
Residual	0.14	17	8.119E-003			
Lack of Fit	*0.14*	*12*	*0.011*	*53.36*	*0.0002*	*significant*
Pure Error	*1.069E-003*	*5*	*2.139E-004*			
Cor Total	0.41	19				

Std. Dev.	0.090	R-Squared	0.6610
Mean	0.23	Adj R-Squared	0.6211
C.V. %	38.76	Pred R-Squared	0.4588
PRESS	0.22	Adeq Precision	11.782



(e) Suppose you had some information that suggests that the separation process malfunctioned during run 7. Delete this run and analyze the data from this experiment again.

Deleting run 7 does not have much impact on the experiment.

Design Expert Output

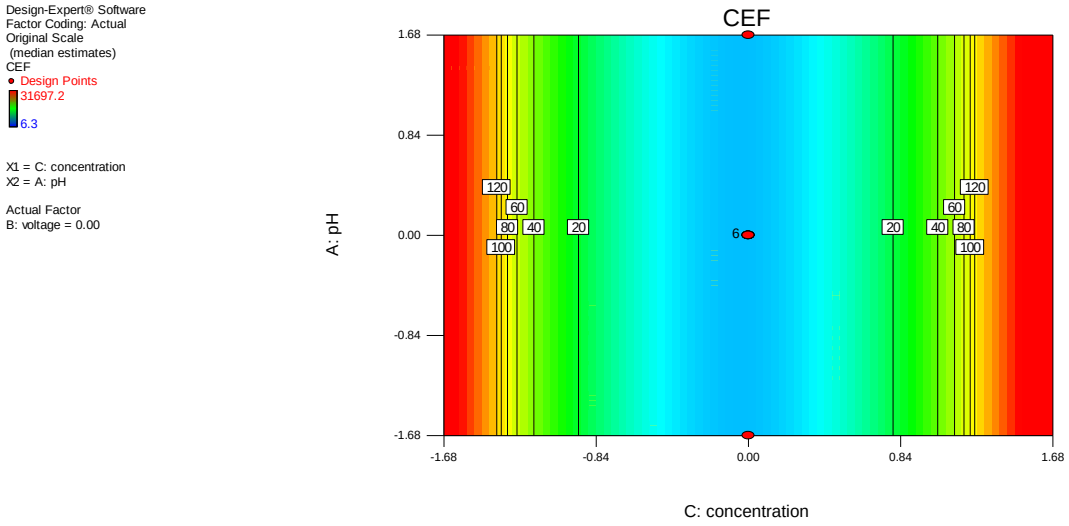
Response 1 CEF
Transform: Inverse Sqrt Constant: 0
ANOVA for Response Surface Reduced Quadratic Model
Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Model	0.25	2	0.12	18.51	< 0.0001	significant
C-conce	*2.928E-004*	*1*	*2.928E-004*	*0.044*	*0.8366*	
C2	*0.25*	*1*	*0.25*	*37.01*	*< 0.0001*	
Residual	0.11	16	6.658E-003			
Lack of Fit	*0.11*	*11*	*9.587E-003*	*44.83*	*0.0003*	*significant*
Pure Error	*1.069E-003*	*5*	*2.139E-004*			
Cor Total	0.35	18				

Std. Dev.	0.082	R-Squared	0.6982
Mean	0.24	Adj R-Squared	0.6605
C.V. %	33.39	Pred R-Squared	0.5392
PRESS	0.16	Adeq Precision	11.599

(f) What conditions would you recommend to minimize CEF?

Run *C* at the zero level minimizes CEF.



11.31. An article in the *Electronic Journal of Biotechnology* ("Optimization of Medium Composition for Transglutaminase Production by a Brazilian Soil *Streptomyces* sp," available at http://www.ejbiotechnology.info/content/vol10/issue4/full/10.index.html) describes the use of designed experiments to improve the medium for cells used in a new microbial source of transglutaminase (MTGase), an enzyme that catalyzes an acyl transfer reaction using peptide-bond glutamine residues as acyl donors and some primary amines as acceptors. Reactions catalyzed by MTGase can be used in food processing. The article describes two phases of experimentation – screening with a fractional factorial and optimization. We will use only the optimization experiment. The design was a central composite design in four factors – $x_1 = KH_2PO_4$, $x_2 = MgSO4\text{-}7H_2O$, x_3 = soybean flower, and x_4 = peptone. MTGase activity is the response, which should be maximized. Table P11.13 contains the design and the response data.

Table P11.13 – The MTGase Optimization Experiment

Std Ord	*X1*	*X2*	*X3*	*X*4	MTGase Activity
1	-1	-1	-1	-1	0.87
2	1	-1	-1	-1	0.74
3	-1	1	-1	-1	0.51
4	1	1	-1	-1	0.99
5	-1	-1	1	-1	0.67
6	1	-1	1	-1	0.72
7	-1	1	1	-1	0.81
8	1	1	1	-1	1.01
9	-1	-1	-1	1	1.33
10	1	-1	-1	1	0.7
11	-1	1	-1	1	0.82
12	1	1	-1	1	0.78
13	-1	-1	1	1	0.36
14	1	-1	1	1	0.23
15	-1	1	1	1	0.21
16	1	1	1	1	0.44
17	-2	0	0	0	0.56
18	2	0	0	0	0.49
19	0	-2	0	0	0.57
20	0	2	0	0	0.81
21	0	0	-2	0	0.9
22	0	0	2	0	0.65
23	0	0	0	-2	0.91
24	0	0	0	2	0.49
25	0	0	0	0	1.43
26	0	0	0	0	1.17
27	0	0	0	0	1.5

(a) Fit a second-order model to the MSGase activity response.

The second-order model includes All 4 quadratic terms A^2, B^2, C^2 and D^2; the two-factor interactions *AB* and *CD* (*AD* is borderline and kept in the model) and the main effects *C* and *D*. *A* and *B* were left in the model for model hierarchy.

Design Expert Output

Response 1 MTGase

ANOVA for Response Surface Reduced Quadratic Model

Analysis of variance table [Partial sum of squares - Type III]

Source	**Sum of Squares**	**df**	**Mean Square**	**F Value**	**p-value Prob > F**	
Model	2.49	11	0.23	10.29	< 0.0001	significant
A-X1	*5.042E-004*	*1*	*5.042E-004*	*0.023*	*0.8817*	
B-X2	*7.704E-003*	*1*	*7.704E-003*	*0.35*	*0.5630*	
C-X3	*0.32*	*1*	*0.32*	*14.73*	*0.0016*	
D-X4	*0.22*	*1*	*0.22*	*9.92*	*0.0066*	
AB	*0.18*	*1*	*0.18*	*8.30*	*0.0114*	
AD	*0.086*	*1*	*0.086*	*3.89*	*0.0674*	
CD	*0.39*	*1*	*0.39*	*17.60*	*0.0008*	
A2	*0.90*	*1*	*0.90*	*41.09*	*< 0.0001*	
B2	*0.58*	*1*	*0.58*	*26.28*	*0.0001*	
C2	*0.44*	*1*	*0.44*	*19.94*	*0.0005*	
D2	*0.56*	*1*	*0.56*	*25.49*	*0.0001*	
Residual	0.33	15	0.022			
Lack of Fit	*0.27*	*13*	*0.021*	*0.69*	*0.7314*	*not significant*
Pure Error	*0.060*	*2*	*0.030*			
Cor Total	2.82	26				

Std. Dev.	0.15	R-Squared	0.8830
Mean	0.77	Adj R-Squared	0.7972
C.V. %	19.38	Pred R-Squared	0.6452
PRESS	1.00	Adeq Precision	12.597

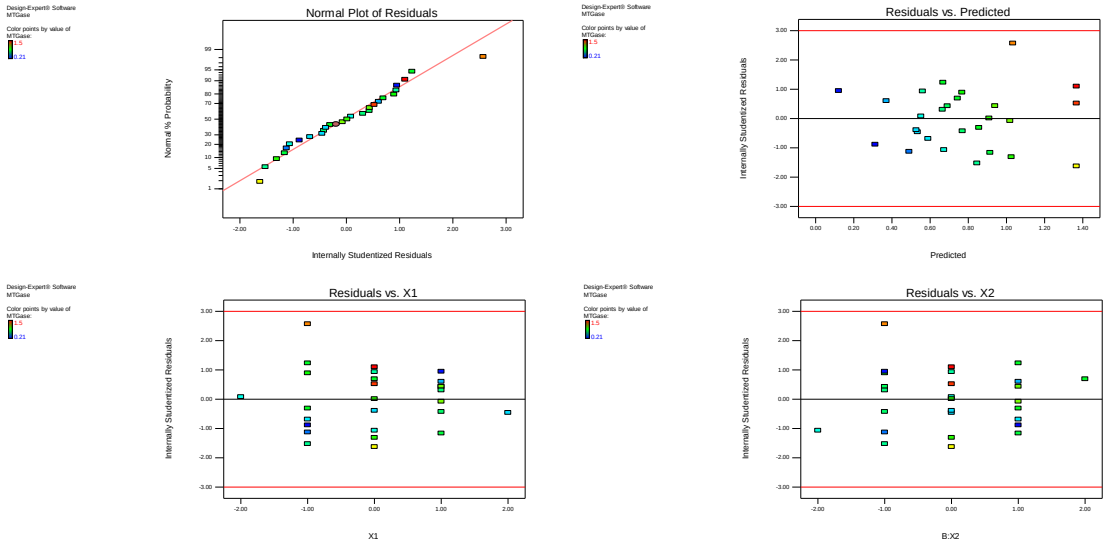
Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	1.37	1	0.086	1.18	1.55	
A-X1	-4.583E-003	1	0.030	-0.069	0.060	1.00
B-X2	0.018	1	0.030	-0.047	0.082	1.00
C-X3	-0.12	1	0.030	-0.18	-0.052	1.00
D-X4	-0.095	1	0.030	-0.16	-0.031	1.00
AB	0.11	1	0.037	0.028	0.19	1.00
AD	-0.073	1	0.037	-0.15	5.940E-003	1.00
CD	-0.16	1	0.037	-0.23	-0.077	1.00
A2	-0.21	1	0.032	-0.27	-0.14	1.25
B2	-0.16	1	0.032	-0.23	-0.096	1.25
C2	-0.14	1	0.032	-0.21	-0.075	1.25
D2	-0.16	1	0.032	-0.23	-0.094	1.25

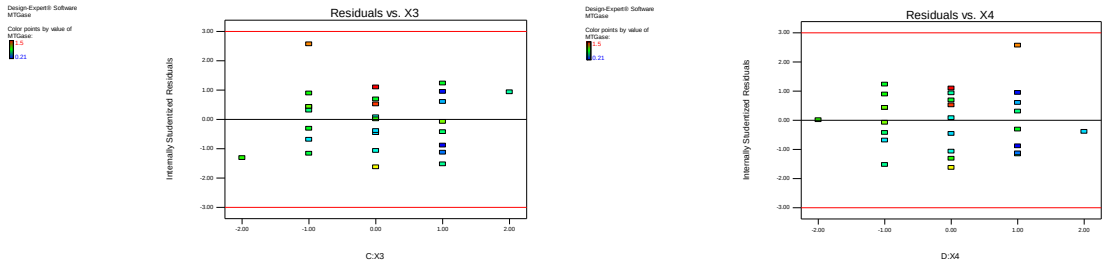
Final Equation in Terms of Actual Factors:

MTGase =	
+1.36667	
-4.58333E-003	* X1
+0.017917	* X2
-0.11625	* X3
-0.095417	* X4
+0.10688	* X1 * X2
-0.073125	* X1 * X4
-0.15563	* X3 * X4
-0.20594	* X12
-0.16469	* X22
-0.14344	* X32
-0.16219	* X42

(b) Analyze the residuals from this model.

Run #9 appears to be an outlier. The residual plots look good for model adequacy.





(c) Recommend operating conditions that maximize MTGase activity.

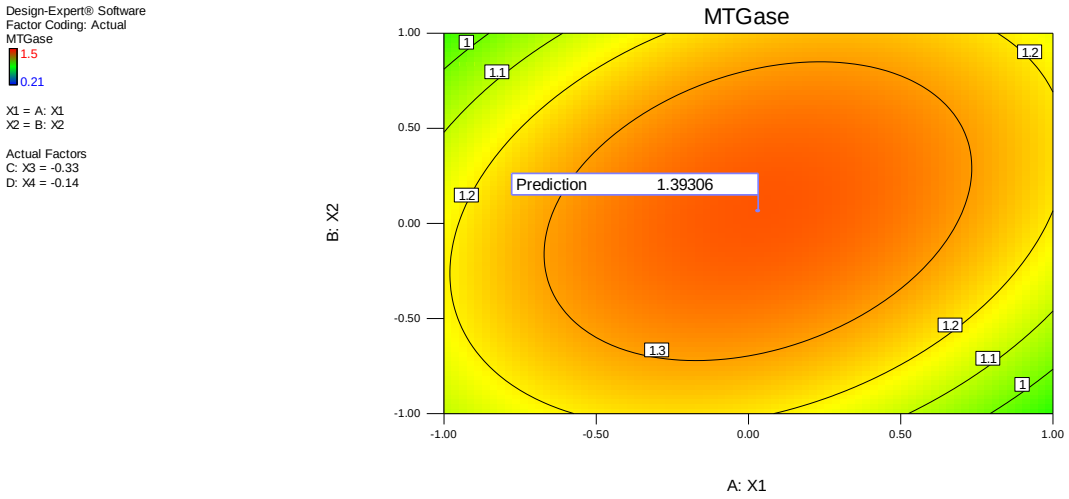
MTGase can be maximized with $X1 = 0.03$, $X2 = 0.07$, $X3 = -0.33$ and $X4 = -0.14$. The MTGase response will be 1.39.

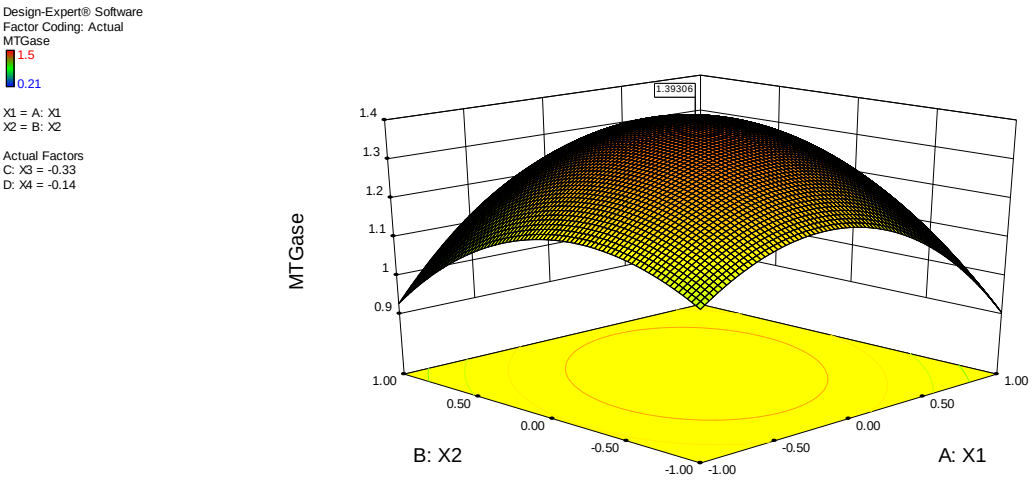
Constraints

Name	Goal	Lower Limit	Upper Limit	Lower Weight	Upper Weight	Importance
A:X1	is in range	-2	2	1	1	3
B:X2	is in range	-2	2	1	1	3
C:X3	is in range	-2	2	1	1	3
D:X4	is in range	-2	2	1	1	3
MTGase	maximize 0.21	1.5	1	1	3	

Solutions

Number	X1	X2	X3	X4	MTGase	Desirability	
1	0.03	0.07	-0.33	-0.14	1.39306	0.917	Selected



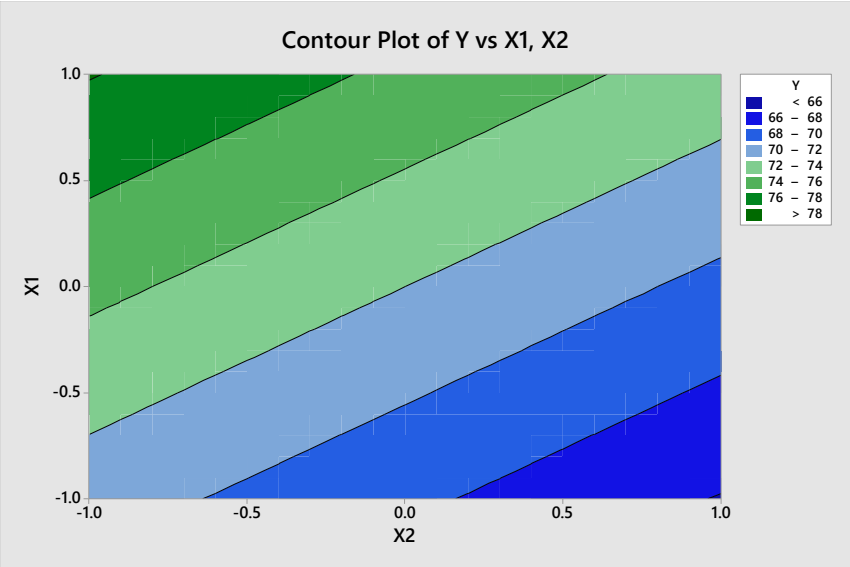


11.32. Given the fitted response function

$$\hat{y} = 72.0 + 3.6x_1 - 2.5x_2$$

which is found to fit well in a localized region of (x_1, x_2).

(a) Plot contours of constant response, y, in the (x_1, x_2) plane.



(b) Calculate and plot the path of steepest ascent generated from this response function. Use x_1 to determine the path, with $\Delta x_1 = 1$.

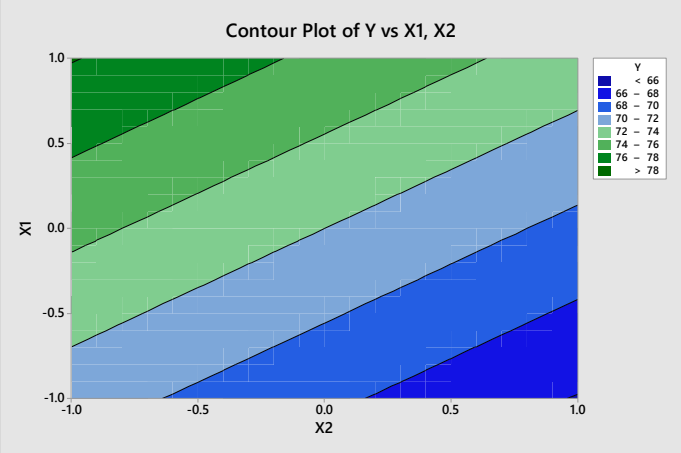
	Coded Variables	
	x_1	x_2
Origin	0	0
Δ	1	-2.5/3.6 = -0.644
Origin +Δ	1	-0.644
Origin +2Δ	2	-1.3888

11.33. It is stated in the text that the development of the path of steepest ascent makes use of the assumption that the model is truly first-order in nature. However, even if there is a modest amount of curvature or interaction in the system, the use of steepest ascent can be extremely useful in determining a future experimental region. Suppose that in a system involving x_1 and x_2 the actual model is given by

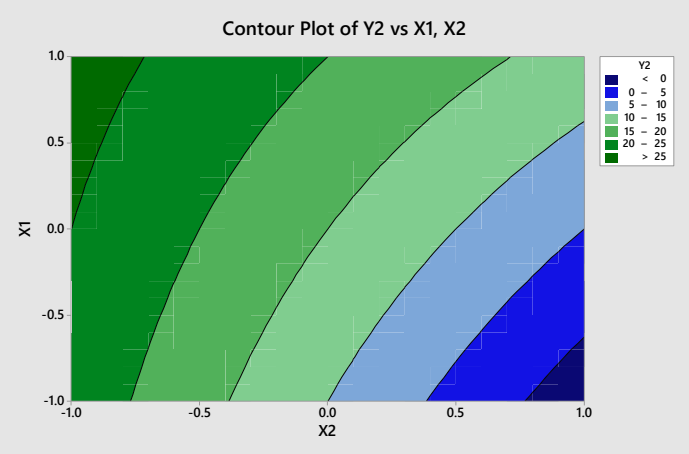
$$E(y) = 14 + 5x_1 - 10x_2 + 3x_1x_2$$

Assume that $x_1$and x_2 are in coded form.

(a) Show a plot of the path of steepest ascent (based on the actual parameters) if the interaction is ignored.



(b) Show a plot of the path of steepest ascent for the model with interaction. Note that this path is not linear.



(c) Comment on the difference in the two paths.

The interaction puts the curvature into the model.

11.34S. Suppose you have fit the following model:

$$\hat{y} = 20 + 10x_1 - 4x_2 + 12x_3$$

(a) The experimenter decides to choose x_1 as the variable to define the step size for the steepest ascent. Do you think that this is a good choice for defining step size? Explain why or why not.

Usually we choose the variable we know the most about or the variable with largest absolute value coefficient. In this case, if we know the most about x_1 then we are good. If not, then x_3 may have been a better choice.

(b) The experimenter decides to make the stop size exactly one coded unit in the x_1 direction. Find the coordinates of this point from the design center on the path of steepest ascent.

	Coded Variables		
	x_1	x_2	x_3
Origin	0	0	0
Λ	1	-4/10 = -0.4	12/10= 1.2
Origin +Δ	1	-0.4	1.2

(c) Suppose that you had used x_3 as the variable to define the direction of steepest ascent. Assuming a step size of one coded unit, find the coordinates of the first step in this direction away from the design center.

	Coded Variables		
	x_1	x_2	x_3
Origin	0	0	0
Δ	10/12 = 0.833	-4/12 = -0.333	1
Origin +Δ	0.833	-0.333	1

(d) Find the distance between the two points found in parts (b) and (c), respectively.

$$\sqrt{(1-0.833)^2 + (-0.4-(-0.33))^2 + (1.2-1)^2} =$$

$$\sqrt{(1.677)^2 + (-0.067)^2 + (0.2)^2} = \sqrt{0.0227 + 0.004 + 0.04} = 0.269$$

The two paths are 0.269 units away from each other. Not very far.

11.35. Any point on the path of steepest ascent is proportional to the unit vector of the first-order regression coefficients.

True False

11.36. Consider the first-order model

$$y = 150 + 20x_1 + 12x_2$$

If the experimenter used one coded unit in the x_1 direction as the stop size, the point $x_1 = 1.5$, $x_2 = 1.0$ is a point on the path of steepest ascent.

True **False**

11.37. A 2^3 factorial design with 4 center points has been used to fit the first-order model

$$\hat{y} = 175 + 25x_1 + 10x_2 - 8x_3$$

The variance of the predicted response at any design point other than the center point is $\frac{11}{24}\sigma^2$

11.38. When conducting steepest ascent experiments, it may be a good idea to replicate runs at some or all of the points along the path.

True False

11.39. If a first-order model with interaction is used, the path of steepest ascent is a straight line.

True **False**

11.40. The coordinates of the points on the path of steepest ascent are proportional to the effect estimates from the first-order model.

True False

11.41. For a first-order model with three variables the variance of the predicted response at the design center (in coded units) is σ^2, if an unreplicated 2^3 design is used to collect the experimental data.

True **False**

11.42. The method of steepest ascent is a gradient procedure.

True False

11.43. Using a second-order model to estimate the underlying relationship is a common approach to approximating the surface.

True False

11.44. The location of the stationary point corresponds to a global maximum or minimum of the surface.

True False

11.45. The existence of a ridge system in the canonical variables can restrict the choice of optimal location.

True **False**

11.46. If the sign of all of the eigenvalues is the same, the response surface will have a global maximum.

True False

11.47. In designing mixture experiments, a good rule of thumb is to assume that at least a second-order mixture model is likely to be adequate.

True False

11.48. The simplex-lattice and simplex-centroid designs have most of the design points on the boundary of the experimental region.

True False

11.49. When axial check blends are added to a simplex-lattice, these blends are as far as possible (Euclidean distance) from the other design points.

True False

11.50. The (3,2) simplex-lattice design is a rotatable design for fitting a second-order mixture model.

True **False**